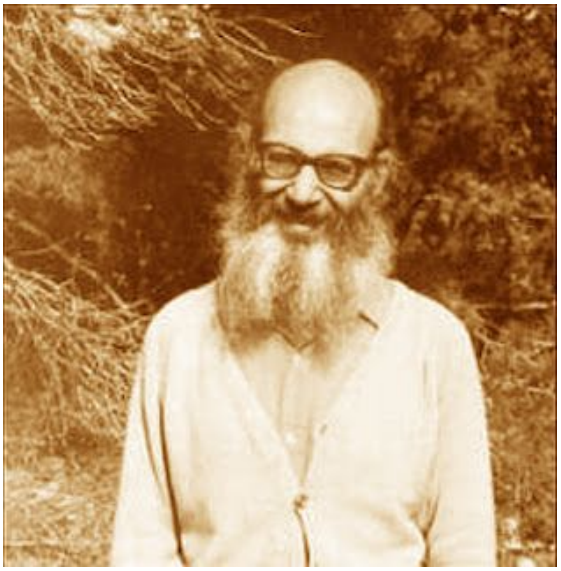
MATH808K
Algebraic K-Theory



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1 Introduction

K-theory is the study of categories of **vector bundles** or similar objects. A vector bundle is a parametrized family of vector spaces: $p : E \to X$ is a vector bundle, X is a topological space. For each $x \in X$, $E_x = p^{-1}(x)$ is a vector space depending "continuously" on $x \in X$. K-theory deals with parametrized linear algebra. Often we don't deal directly with geometry, but with rings

Theorem 1.1 (Swan-Serre) There is an equivalence of cateogories between vector bundles over X and finitely generated projective modules over an associated ring of functions on X. Here are 3 categories in which this works

1. X compact Hausdorff, $R = C(X)$ the continuous $\mathbb{F}$-valued function on X. Here $\mathbb{F}$ could be $\mathbb{R}, \mathbb{C}$ or $\mathbb{H}$
2. X affine variety over a field k, $R = \mathcal{O}(X)$ is the ring of regular functions. If X is projective, it is more complicated
3. X stein manifold(holomorphic submanifold of $\mathbb{C}^n$), $R = \mathcal{O}(X)$ holomorphic functions on X, category of vector bundles is holomorphic category

2 Projective modules

In this course, a ring almost always have units but not necessarily commutative

Definition 2.1 R is a ring with unit. A free R-module is one isomorphic to R^I, I is some index set. A finitely generated free R-module is one isomorphic to R^n. R is said to have the *invariant basis property* if $R^n \cong R^m \Rightarrow n = m$. Note that this is always true if R is commutative(reason: true for fields, and if R is commutative, $k = R/m$ is field, $R \otimes k$ is a vector space over k)

Exercise 2.2 (Counter-example) k is a field, $R = \mathrm{End}_k(k^\infty)$ doesn't have the invariant basis property, as $R \cong R^2$. Idea: $k^\infty \oplus k^\infty \cong k^\infty$

Solution. R corresponds to a matrix with rows and columns ranging in $\mathbb{N}$, and each column has all but finitely many non-zero elements, and we can decompose any matrix to the direct sum of two matrices with even columns or odd columns

$$\begin{bmatrix} \circ & \times & \circ & \times & \circ & \times & \cdots \\ \circ & \times & \circ & \times & \circ & \times & \cdots \\ \circ & \times & \circ & \times & \circ & \times & \cdots \\ \circ & \times & \circ & \times & \circ & \times & \cdots \\ \circ & \times & \circ & \times & \circ & \times & \cdots \\ \circ & \times & \circ & \times & \circ & \times & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

□

Theorem 2.3 R is ring, P is an R-module. The following are equivalent

1. P is a direct summand in a free R-module, i.e. $F \cong P \oplus Q$ for some free R-module F
2. $\mathrm{Hom}_R(P, -)$ is an exact functor
3. P has the property that if $\phi : M \to N$ is a surjective R-module map and we are given $\alpha : P \to N$, there exists $\beta : P \to M$ such that $\alpha = \phi \circ \beta$

An R-module with these 3 equivalent conditions is called *projective*

Proof.

□

The first invariant of K-theory is $K_0(R)$, then **Grothendieck group** of finitely generated projective modules over R. If P and Q are finitely generated projecive R modules, we can "add" by taking direct sum, but not subtract. $K_0(R)$ is the group with generators $[P]$, P finitely generated R-module with relations $[P] = [Q]$ if $P \cong Q$. We build in the relation $[P] + [Q] = [P \oplus Q]$. Note that every element of $K_0(R)$ is of the form $[P] - [Q]$ for some P, Q, $[P] - [Q] = [P'] - [Q'] \Leftrightarrow P \oplus Q' \oplus S \cong R' \oplus Q \oplus S$ for some S. In general, "addition" of projective modules does not have the cancellation property, just as addition of vector bundles does not

Example 2.4 TS^2 is not free since $\chi(S^2) = 2 \neq 0$

Fact 2.5 (reference: Hatcher's K-theory book)

1. Any vector bundle (by definition) is locally trivial, then rank $X \to \mathbb{N}$ is continuous, hence locally constant
2. Any vector bundle can be equipped with a metric, i.e. a family of inner products varying continuously with $x \in X$. (Construction: use local triviality and patch with partition of unity)
3. Any vector bundle can be embedded into a trivial vector bundle $X \times \mathbb{F}^n$ for n large enough. (Use local triviality and partition of unity)
4. 2+3 $\Rightarrow$ Any vector bundle is a direct summand in a trivial bundle

Proof of Theorem 1.1. Send $p : E \to X$ to the set of sections $\Gamma(E)$, then $\Gamma(E)$ is a $\mathcal{O}(X)$-module, from above, $\Gamma(E)$ is finitely generated and projective. The rest is formal

□

Example 2.6 Observation: Any vector bundle over $S^n, n \geq 1$ is obtained by gluing("clutching"): two trivial vector bundles over the upper and lower hemispheres via a map $S^{n-1} \to \mathrm{GL}\ (k, \mathbb{F})$. This is because any vector bundle over a contractible space is trivial, so

$$\mathrm{Vect}^k_{\mathbb{F}}(S^n) \cong [S^{n-1}, \mathrm{GL}\ (k, \mathbb{F})] \cong \begin{cases} \pi_{n-1}(O(k)) & \mathbb{F} = \mathbb{R} \\ \pi_{n-1}(U(k)) & \mathbb{F} = \mathbb{C} \\ \pi_{n-1}(\mathrm{Sp}\ (k)) & \mathbb{F} = \mathbb{H} \end{cases}$$

$X = S^2$, $\mathbb{F} = \mathbb{R}$, what is the classification of rank n vector bundles over X? We see that rank k vector bundles over S^2 are classified by $\pi_1(O(k))$, since S^1 is connected, any map $S^1 \to O(k)$ lies in a single component, both isomorphic to SO (k), for $k \geq 3$, SO (k) is a simple Lie group and $\pi_1(\mathrm{SO}\ (k)) \cong \mathbb{Z}/2$ for $k \geq 3$(except SO (4) is only semi-simple with two cover?)

Note. $SU(2)$ as quaternions acts on S^3, gives a double cover of SO (3)

Implication for K-theory: The stable isomorphic classes of vector bundles E over S^2 is characterized by

$$\begin{cases} \text{rank} \in \mathbb{N} \\ \text{Stiefel-Whitney number} = \langle w_2(E), [S^2] \rangle \in \mathbb{Z}/2 \end{cases}$$

Similar analysis holds for S^n

$$\begin{cases} \text{rank} \in \mathbb{N} \\ \text{something in}\ \ \pi_{n-1}(\mathrm{SO}) \end{cases}$$

Here $\pi_{n-1}(\mathrm{SO}) = \pi_{n-1}(\mathrm{SO}\ (\infty)) = \lim_{\underset{k}{\to}} \pi_{n-1}(\mathrm{SO}\ (k))$, $\pi_{n-1}(\mathrm{SO}\ (k))$ stablizes as $k \to \infty$

Theorem 2.7 (Bott periodicity theorem)

$$\pi_{n-1}(\mathrm{SO}) = \begin{cases} \mathbb{Z}, & n \ \text{ is a multiple of } \ 4 \\ \mathbb{Z}/2, & n \equiv 1, 2 \mathrm{mod}\ 8 \\ 0, & \text{otherwise} \end{cases}$$

Lessons form this example: stable classification is much easier than the unstable classification. A stably trivial bundle need not to be trivial. These lessons carry over to the purely algebraic setting of projective modules over a ring. To get a corresponding example with projective modules over a Noetherian commutative ring, take $R = \mathbb{R}[x, y, z]/(x^2 + y^2 + z^2 - 1)$, Spec R is an "algebraic model" for S^2. Our non-trivial but stably trivial vector bundle can be constructed as $\{(x, y, z, u, v, w) \mid x^2 + y^2 + z^2 = 1, xu + yv + zw = 0\}$. This is an algebraic model for the tangent bundle of S^2

3 Homotopy Invariance

Theorem 3.1 The classification of the topological vector bundles over a compact Hausdorff space X is homotopy invariant. In other words, if $f, g : X \to Y$ are maps of compact spaces and E is an $\mathbb{F}$ bundle over Y, then $f \simeq g \Rightarrow f^*(E) \cong g^*(E)$

Corollary 3.2 Every vector bundle over a contractible space is trivial

Theorem 3.3 A is a unital Banach algebra(For application, $A = C(X, M_n(\mathbb{F}))$. Let Idem A be the set of idempotents in $A(x^2 = x)$. If $e, f \in$ Idem A lies in the same component, then they are conjugate under $\mathrm{GL}_1(A)$

Proof. It's enough to show that if $e, f \in$ Idem A are sufficiently close in norm, then e, f are conjugate. Suppose e, f are close and let $a = e + f - 1 \in A$, then a^2 is close to $(2e - 1)^2 = 1$, so a^2 is invertible, thus a is invertible. $ae = fa$ since $ae = (e + f - 1)e = fe$, $fa = f(e + f - 1) = fe$, thus $aea^{-1} = f$

□

Proof Theorem 3.1. Embed E as a direct summand in a trivial bundle of rank n, then $f^*(E), g^*(E)$ are obtained by projecting down from $X \times \mathbb{F}^n$ via homotopic idempotents in $C(X, M_n(\mathbb{F}))$

□

4 Projective modules over a local ring

Definition 4.1 R is a ring with unit, R is called **local** if the non-invertible elements in R constitute a 2-sided ideal m. Obviously m is the unique maximal 2-sided ideal

Caution: In the non-commutative case, having a unique maximal 2-sided ideal is not good enough! Since $M_n(\mathbb{F})$ has this property for $\mathbb{F}$ a field, and this ring is not local

Note: If R is local and $x \in R$ has a left inverse, then it also has a right inverse. Suppose $ax = 1$, then $ax \notin m$, so $x \notin m$, so x is invertible

Fact 4.2

1. If R is local with maximal ideal m and $x \in m$, then $1 + x$ is invertible. If not, then $1 + x \in m \Rightarrow 1 \in m$ which is a contradiction
2. (Nakayama's lemma) R is a local ring with maximal ideal m, and let m be a finitely generated R-module, if $mM = M$, then $M = 0$. Let $M = Rx_1 + \cdots + Rx_n$ such that n is minimal, then since $mM = M$, $x_n = r_1x_1 + \cdots + r_nx_n$ with $r_j \in m$, now $(1 - r_n)x_n = r_1x_1 + \cdots + r_{n-1}x_{n-1}$, but $1 - r_n$ is invertible, we can divide to get x_n, contradicting the minimality unless $n = 0$, i.e. $M = 0$

Corollary 4.3 (Corollary of Nakayama's lemma) M is a finitely generated R-module, $x_1, \cdots, x_k \in M$

1. If $\dot{x}_1, \cdots, \dot{x}_k$ generate M/mM, then it generate M
2. If $\dot{x}_1, \cdots, \dot{x}_k$ is a free basis for M/mM, then it is a free basis for M

Proof. Consider $N = M/(Rx_1 + \cdots + Rx_k)$, this shows that $x_1, \cdots, x_k \in M$ generates M. To show this is a basis, consider

□

Theorem 4.4 Let R be a local ring, M a finitely generated projective R-module, then M is free

Proof. $M \oplus N \cong R^n$. $m(M \oplus N) = m^n$, so $(R/m)M$ is a direct summand in $(R/m)^n$, but R/m is a division ring, so $(R/m)M \cong (R/m)^k$ for some $0 \leq k \leq n$, and $(R/m)N = (R/m)^{n-k}$. Let $\dot{x}_1, \cdots, \dot{x}_k$ be a free basis for $(R/m)M \cong M/mM$ and extend it to a free basis by adding $\dot{x}_{k+1}, \cdots, \dot{x}_n$ for $(R/m)N$, pull these back to $x_1, \cdots, x_k \in M$ and $x_{k+1}, \cdots, x_n \in N$. $M = Rx_1 + \cdots + Rx_k$ by Nakayama's lemma. Since $x_1, \cdots, x_n$ is another generating set for R^n with n elements, writing the the matrix of x_i's and e_i's gives the linear independence

□

Corollary 4.5 For R is a local ring, $K_0(R) = \mathbb{Z}$, with the class of a projective module given by its rank(this is only stable case, note that the theorem actually prove the non-stable case, which is more general)

5 Projective modules versus vector bundles

We discussed the Swan-Serre theorem, that if E is a vector bundle over a compact Hausdorff space X, then $\Gamma(X, E)$ is a finitely generated projective $C(X)$-module. This suggests a deeper connection between properties of vector bundles and finitely generated projective moduels. A key aspect of vector bundles is that they are **locally trivial**. The algebraic analogue of this is **local freeness**

Theorem 5.1 (Lemma 2.4, Weibel) R is a commutative ring, M is a finitely generated R-module. The following are equivalent

1. M is projective
2. M is locally free, i.e. for each prime ideal p, $\exists s \in R \setminus p$ such that $M[1/s]$ is free of finite rangk over $R[1/s]$
3. For each prime ideal p of M, M_p is a free R_p-module, and M is finitely presented, i.e. there is an exact sequence $R^m \to R^n \to M \to 0$(If R is Noetherian, then finitely generated implies finitely presented)

Proof. 1⇒2,3: Assume M is projective, $R^n \cong M \oplus P$, then M is finitely presented, i.e. $R^n \to R^n \to M \to 0$. Also $R_p^n \cong M_p \oplus P_p$ over teh local ring R_p, by Theorem 5.1, M_p is finitely generated and free over R_p. So after inverting things in $R \setminus p$, we get a free module. But we really only have to invert finitely many elements since M is finitely presented, which is the same as inverting their product, which is a single $s \in M \setminus p$, so $M[1/s]$ is free of finite rangk over $R[1/s]$
3⇒1: Since M is finitely presented, we have an exact sequence $R^m \to R^n \xrightarrow{\epsilon} M \to 0$. To show M is projective, it suffices to show that $\epsilon^* : \text{Hom}\,(M, R^n) \to \text{Hom}\,(M, M)$ is surjective, since id_M then has a preimage. Given a prime ideal p, M_p is free over R_p, so ϵ_p^* is split surjective, so is ϵ^*, since $\text{Hom}\,(M, M)/\,\text{Hom}\,(M, R^n)$ is 0 after localizing at any prime ideal, hence 0

□

So at least over a Noetherian commutative ring, M finitely generated projective ⇔ M_p finitely generated free for each $p \in \text{Spec}\ R$. This suggests(and we used this to study vector bundles over spheres) that to construct interesting projective modules, we get them by patching(clutching) finitely generated free modules over different open sets

Example 5.2 (Milnor square, his book on algebraic K-theory) Suppose $f : R \to S$ is a map of rings, $I \cong f(I)$, then R is the puback of the diagram

$$\begin{array}{ccc} R & \xrightarrow{f} & S \\ \downarrow & & \downarrow \\ R/I & \xrightarrow{f} & S/I \end{array}$$

Think of S as R/J, so we are patching over I and J

Example 5.3 (Topological example) $R = C(S^2)$, I is the ideal of functions which vanish on the open upper hemisphere and tend to zero along the equator. Then $R/I \cong C(D^2_-)$, continuous functions on the closed lower hemisphere. Let S be the continuous functions on the closed upper hemisphere $C(D^2_+)$, $R \to S$ is just restriction S and R/I are both of the form $C(D^2)$, and since D^2 is contractible, every vector bundle on D^2 is trivial, and so every finitely generated projective S-module or R/I-module is free. But S^2 is contractible, we want to describe fintiely generated projective R-modules in terms of patching of trivial bundles along the equator

Theorem 5.4 (Milnor patching theorem) In a Milnor square

1. An R-moudle P obtained by patching a finitely generated projective S-module P_1 and a finitely generated projective R/I-module P_2 is finitely genereated and projective
2. In the situation of 1, $S \otimes_R P \cong P_1$ and $P/IP \cong P \otimes_R R/I \cong P_2$
3. (important)Every finitely generated projective R-mod arises this way
4. Suppose P is obtained by patching S^n and $(R/I)^n$ via $g \in \mathrm{GL}_n(S/I)$, and Q is obtained similarly via $g^{-1} \in \mathrm{GL}_n(S/I)$, Then $P \oplus Q \cong R^{2n}$
5. More generally, if P is obtained by patching S^n and $(R/I)^n$ via $g \in \mathrm{GL}_n(S/I)$, and Q is obtained by patching S^n and $(R/I)^n$ via h, then $P \oplus Q \cong$ patching of S^n and $(R/I)^n$ via $gh \oplus R^n$

Proof. Translate to algebra the idea of gluing vector bundles when $X = X_1 \cup_Y X_2$, here X_1, X_2 are closed subsets of a compact Hausdorff sopace X_1 and $Y = X_1 \cap X_2$. Alternatively, the Milnor patching theorem is a kind of "algebraic Mayer-Vietoris"

proof of 3: Recall R is embedded in $S \oplus R/I$. Let P be a finitely generated projective R-module Let $P_1 = S \otimes_R P$ which is finitely generated projective S-module, let $P_2 = P/IP$ is a finitely generated projective R/I-module. We want to show that P is obtained by patching P_1 and P_2. We have an exact sequence of R-modules

$$0 \to R \to S \oplus R/I \to S/I \to 0$$

$\mathrm{Hom}_R(P, -)$ and $- \otimes_R P$ is exact since P is projective, so we have exact sequence

$$0 \to P \to P_1 \oplus P_2 \to S/I \otimes_R P$$

which says that P is patched from P_1 and P_2 $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \simeq \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ through homotopy $\begin{bmatrix} 1-t & -t \\ t & 1-t \end{bmatrix}$ which has determinant $(1-t)^2 + t^2 > 0$, hence $\begin{bmatrix} g & 0 \\ 0 & g^{-1} \end{bmatrix} = \begin{bmatrix} 1 & g \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -g^{-1} & 1 \end{bmatrix} \begin{bmatrix} 1 & g \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ and $\begin{bmatrix} g & 0 \\ 0 & h \end{bmatrix} = \begin{bmatrix} g & 0 \\ 0 & g^{-1} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & gh \end{bmatrix}$, here $\begin{bmatrix} 1 & g \\ 0 & 1 \end{bmatrix} \simeq \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ through homotopy $\begin{bmatrix} 1 & tg \\ 0 & 1 \end{bmatrix}$

□

6 Line bundles and Picard Groups

We assume in this section R is commutative. We showed that a finitely generated R-module P is projective exactly when it's locally free. We call it(by analogy with topology) a **line bundle** if P

is **locally free of rank 1**. Commutativity of R enables you to tensor two R-modules together to get a new one. This preserves the property of being a line bundle. So we can define the **Picard group** Pic (R) of R, to be the abelian group of line bundles over R up to isomorphism with tensor product as the group operation and $[R]$ as the identity element. The dual line bundle $P^\vee = \mathrm{Hom}_R(P, R)$, and $P \otimes_R P^\vee \to R$ is an isomorphism, so $[P^\vee] = [P]^{-1}$. For topological line dunles over $\mathbb{C}$, the dual line bundle can also be identified with the complex conjuate bundle. The dual of vector bundle $E \to X$ is the bundle whose fibers are the dual vector spaces. Over $\mathbb{C}$, $\mathbb{C}^* \cong \overline{\mathbb{C}}$ via the usual inner product

Example 6.1

1. $R = C^{\mathbb{R}}(X)$, X compact Hausdorff. Pic $(R) \cong H^1(X, \mathbb{Z}/2)$
2. $R = C^{\mathbb{C}}(X)$, X compact Hausdorff. Pic $(R) \cong H^1(X, \mathbb{Z})$

Theorem 6.2 R is a commutative ring, if two line bundle are stably isomorphic, then they are actually isomorphic

6.1 Rank and Dimension

Here are some more properties of rank of finitely generated projective modules over a commutative ring R. They are algebraic analogues of some theorems in topology

Theorem 6.3 If M is a manifold of dimension n and E is a vector bundle of rank k which is stably trivial, then E is trivial provided $\begin{cases} k>n & \mathbb{F}=\mathbb{R} \\ k>(n+1)/2 & \mathbb{F}=\mathbb{C} \end{cases}$

Example 6.4 The tangent bundle of S^2 is stably trivial but not trivial, in this case, $n = 2, k = 2, \mathbb{F} = \mathbb{R}$. But every stably trivial 3 plane bundle over S^2 is trivial

Proof. There is a fibration $O(n-1) \to O(n) \to S^{n-1}$, we have $\pi_{j+1}(S^{n-1}) \to \pi_j(O(n)) \to \pi_j(O(n-1)) \to \pi_j(S^{n-1})$. The map $\pi_j(O(n-1)) \to \pi_j(O(n))$ is sujective if $j < n-1$. an iso if $j < n-2$. and we can decompose M as $(n-1)$-skeleton$\cup\, D^n$ via gluing along the $(n-1)$-skeleton

Similarly with $U(n-1) \to U(n) \to S^{2n-1}$, which gives $\pi_j(U(n-1)) \to \pi_j(U(n))$ surjective if $j < 2n-1$, an iso if $j < 2n-2$

□

6.2 The rank function

R is a commutative ring, we get a notion of local rank function of a finitely generated projective R-mod P, Spec $(R) \to \mathbb{N}, p \mapsto \text{rank } (P_p)$.

Lemma 6.5 The rank funciton is continuous in Zariski topology

Proof. p is contained in a maximal ideal m, rank Spec $R[1/s] =$ rank $(P_p) =$ rank (P_m). We get a partition of Spec R into disjoint open sets

□

Theorem 6.6 (Stable rank theorem, Bass-Serre, Bass's book on algebraic K theory, not easy) R commutative and Noetherian of dimension d. If R has a finitely projective module P of constant rank k. then if $k > d$

1. $P = P_0 \oplus R^{k-d}$ for some projective module P_0 of rank d
2. P is stably isomorphic to P' of same rank $\Rightarrow P \cong P'$
3. P stably free $\Rightarrow P$ is free

Corollary 6.7 If R is a commutative ring with no idempotents other than 0 and 1, then every finitely generated projective R-module has constant rank

Proof. Spec R is connected iff R has no non-trivial idempotent. And an integral domain is connected(if x is an idempotent, so is $1-x$, and $x(1-x)=0$)

□

6.3 Back to line bundles

Let R be a commutative ring, P finitely generated projective R-module. We can form new modules $\bigwedge^j P$. If P has constant rank k, then $\bigwedge^j P$ is projective of rank $\binom{k}{j}$. So we can define the determinant of P, $\det(P) = \bigwedge^k P$,$k = \text{rank } P$. If R has no non-trivial idempotents, the determinant module is defined for all finitely generated projective modules

1. $\det P$ is a line bundle
2. $\det(P \oplus Q) = \det P \otimes \det Q$

From these facts we deduce

Theorem 6.8 R is a commutative ring, if two line bundles are stably isomorphic iff isomorphic, i.e. define the same class in Pic (R)

Proof. Assume $P \oplus R^n \cong P' \oplus R^n$, then $P \cong \det P \otimes R = \det(P \oplus R^n) \cong \det(P' \oplus R^n) = \det P' \otimes R \cong P'$

□

Theorem 6.9 R commutative Noetherian of dimension 1(in particular, Dedekind domain: Noetherian integral domain which is integrally closed in the field of fractions). Then every finitely generated projective R-module of constant rank is of the form $P \oplus R^{k-1}$ for some k and for some line bundle P, namely the determinant bundle of the module. Thus finitely generated projective R-module are completely classified by rank and determinant

Proof. Just use stable rank theorem

□

Corollary 6.10 If R is a commutative Noetherian integral domain of dimension 1, then $K_0(R) \cong \mathbb{Z} \oplus \text{Pic } (R)$, $[P] \mapsto (\text{rank } P, \det P)$

Note $\det(P \oplus Q) = \det P \otimes \det Q$ says it is a homomorphism in Pic, rank adding says it is a homomorphism in $\mathbb{Z}$

7 Ideal classes and Picard groups

R commutative integral domain, F its field of fractions. A fractional ideal of R is a non-zero R -submodule I of F, is such that $xI \subseteq R$ for some $x \in F^\times$ (i.e. we can clear denominators). I is invertible if $IJ = R$ for some other fraction ideal J. Principal fractional ideals are Rx for $x \in F^\times$. Principal fractional ideals are invertible. The invertible fractional ideals form the Cartier group Cart (R) of Cartier divisors under multiplication

Proposition 7.1 The line bundles over R are the same as invertible ideals up to isomorphism. If I, J are fractional ideals and I is invertible, then $I \otimes_R J \cong IJ$. There is an exact sequence

$$1 \to R^\times \to F^\times \xrightarrow{\text{div}} \text{Cart } (R) \to \text{Pic } (R) \to 0$$

Proof. Suppose I is an invertible fractional ideal, $IJ = R$, then there exist $x_i \in I, y_i \in J$, $\sum_{i=1}^n x_i y_i = 1$, $\{x_i\}$ define homomorphisms $R^n \to I$, $\{y_i\}$'s define homomorphisms $I \to R^n$. $I \to R^n \to I$ is the identity, so I is projective, and has rank one because $I \otimes_R F = F$, so I is a line bundle. Conversely, suppose L is a line bundle, we get an embedding $L = R \otimes_R L \subseteq F \otimes_R L = F$, L is isomorphic to a nonzero R submodule of F, i.e. a fractional ideal. Since I is finitely generated and projective, we have $R \cong L \otimes_R L^\vee$, hence L is invertible

□

Equivalent definition for Dedekind domain: Integral domain for which every fractional ideal is invertible

Theorem 7.2 R Dedekind domain, every finitely generated torsion-free R-module M is projective and isomorphic to $I \otimes R^{n-1}$ for some fractional ideal I

Proof. Induction on $\text{rank } M = \dim_F(F \otimes_R M)$. M torsion-free means M embeds into $F \otimes_R M$, so if $\text{rank } M = 0$, $M = 0$. Suppose $\text{rank } M = n$, M embeds into F^n and then projects to the last factor F, image can't be zero, so we have $0 \to M_0 \to M \to I \to 0$ with I fractional ideal projective, so $M = M_0 \oplus I$, by induction, $M = I \oplus J \oplus R^{n-2}$. $I \oplus J = IJ \oplus R$ since they have the same determinant

□

7.1 Applications

1. If R is a PID, then $\text{Pic } (R) = 0$, $1 \to R^\times \to F^\times \to \text{Cart } (R) \to 1$ is exact
   1. $R = \mathbb{Z}, F = \mathbb{Q}$, get

      $$1 \to \mathbb{Z}^\times \to \mathbb{Q}^\times \to \mathbb{Q}^\times/\{\pm 1\} = \mathbb{Q}^\times_{>0} \to 1$$

   2. $R = \mathbb{Z}[i], F = \mathbb{Q}[i]$, $R^\times = \{\pm 1, \pm i\}$
   3. $R = k[t], F = k(t)$, $R^\times = k^\times$
2. F be a number field, $R = \mathcal{O}_F$. Then R is a Dedekind domain, the $\text{Pic } (R)$ is finite, and $R^\times$ is finitely generated, in fact, $\text{rank } (R^\times) = r_1 + r_2 - 1$. Most of the time, R is not a PID
3. k is a field, C is a smooth affine curve over k, R is the ring of regular functions on C, then R is a Dedekind domain. In the case $k = \mathbb{C}$, $\text{Pic } (R)$ is the group of algebraic line bundles. $\text{Cart } (R)$ indeed corresponds to the group of divisors(finite linear combinations of points)

8 Mayer-Vietoris for Picard groups

Recall that if R is a commutative ring, $\text{Pic } (R)$ is the group of isomorphism classes of line bundles over R. Thiss can sometimes be computed by "patching" or "Mayer-Vietoris"

Theorem 8.1 (In Weibel I.3) In a Milnor square

$$\begin{array}{ccc} R & \xrightarrow{f} & S \\ \downarrow & & \downarrow \\ R/I & \xrightarrow{\dot{f}} & S/I \end{array}$$

we get an eaxt sequence

$$1 \to R^\times \xrightarrow{\Delta} S^\times \times (R/I)^\times \to (S/I)^\times \xrightarrow{\partial} \text{Pic } (R) \to \text{Pic } (S) \times \text{Pic } (R/I) \xrightarrow{\text{ratio}} \text{Pic } (S/I)$$

Here Δ is the diagonal map and the maps $S^\times \times (R/I)^\times \to (S/I)^\times$, $\text{Pic } (S) \times \text{Pic } (R/I) \to \text{Pic } (S/I)$ are the "ratio maps" $(x, y) \mapsto \frac{\dot{x}}{\dot{f}(y)}$ (with multiplicative notation). Note that this is highly analogous to Mayer-Vietoris in topology. The connecting homomorphism ∂ is given by the gluing map.

9 Classification of topological vector bundles and characteristic classes

Let X be a compact Hausdorff space, $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. Let $E \to X$ be a rank k vector bundle. Recall that E embeds in a trivial rank n bundle for some n sufficiently large, then at each point $x \in X$, we have an associated fiber $E_x \subseteq \mathbb{F}^n$. thus we get a continuous map $X \to \mathrm{Gr}_{\mathbb{F}}(k, n)$

$$\mathrm{Gr}_{\mathbb{F}}(k,n) = \begin{cases} O(n)/O(k) \times O(n-k) & \mathbb{F} = \mathbb{R} \\ U(n)/U(k) \times U(n-k) & \mathbb{F} = \mathbb{C} \end{cases}$$

Over $\mathrm{Gr}_{\mathbb{F}}(k, n)$, we have a canonical or universal bundle of rank k. Now we get classifying space $BO(k) = \lim\limits_{\substack{\to \\ n\to\infty}} \mathrm{Gr}_{\mathbb{R}}(k, n)$ and $BU(k) = \lim\limits_{\substack{\to \\ n\to\infty}} \mathrm{Gr}_{\mathbb{C}}(k, n)$, $BO(1) = \mathbb{RP}^\infty$, $BU(1) = \mathbb{CP}^\infty$

Theorem 9.1 (Classification theorem) X is a compact Hausdorff space, there is a natural bijection

$$\{\text{isomorphism class of rank } \ k \ \text{ real vector bundle over } \ X\} \leftrightarrow [X, BO(k)]$$

$$\{\text{isomorphism class of rank } \ k \ \text{ complex vector bundle over } \ X\} \leftrightarrow [X, BU(k)]$$

Given a map $X \to BO(k)$, the associated vector bundle is just the pullback. Note that since X is compact, the image lands in $\mathrm{Gr}\ (k, n)$ for some n

Proof. We saw the every vector bundle arises this way, we also showed that homotopic maps give isomorphic vector bundles. It remains to show that isomorphic vector bundles come from homotopic maps. Suppose $E \cong F$ are rank k vector bundles, both embedded in a trivial n-dimensional vector bundle. Taking n large enough we can construct ϕ as a map $X \to \mathrm{GL}_n(\mathbb{F})$, $\phi(x) : E_x \to F_x$, sending E_x and $E_x^\perp$ to F_x and $F_x^\perp$ isomorphically. We want to show that maps classifying E and F are homotopic. Trick: double n, and replace ϕ by $\begin{bmatrix} \phi & \\ & \phi^{-1} \end{bmatrix}$ This map can be homotoped to the identity. This shows the classifying maps to $\mathrm{Gr}_{\mathbb{F}}(k, 2n)$ are homotopic

□

9.1 Characteristic classes

It turns out that $H^*(BO(k), \mathbb{Z}/2)$ and $H^*(BU(k), \mathbb{Z})$ are polynomial rings on k variables(prove this by induction on k using the fibrations $O(k-1) \to O(k) \to S^{k-1}$, $U(k-1) \to U(k) \to S^{2k-1}$). From this, one gets the following result

Theorem 9.2 (Existence and uniqueness theorem for characteristic classes, ref: see Hatcher on vector bundles and K theory) There are unique classes $w_j \in H^j(BO(k), \mathbb{Z}/2)$ with the following properties

1. $H^*(BO(k), \mathbb{Z}/2) \cong \mathbb{Z}/2[w_1, \cdots, w_k]$
2. If L is a real line bundle over X, w_1 pulls back to the element of $H^1(X, \mathbb{Z}/2) = \text{Hom}\ (\pi_1(X), \mathbb{Z}/2)$ which classifies the line bundles, as $\text{Hom}\ (\pi_1(X), \mathbb{Z}/2)$ corresponds to the two to one coverings $\tilde{X}$, and $L = \tilde{X} \times_{\pi_1(X)} \mathbb{R}$
3. For arbitrary real vector bundles E of rank k over X, we define $w(E)$, called the total Stiefel-Whitney class of E, to be $1 + f^*(w_1) + f^*(w_2) + ...$ which has the property that $w(E) = w(F)$ if $E \cong F$, $w(E)w(F) = w(E \oplus F)$, $w(\text{trivial bundle}) = 1$. These imply that w is a stable invariant

There are unique classes $c_j \in H^j(BU(k), \mathbb{Z})$ with the following properties

1. $H^*(BU(k), \mathbb{Z}) \cong \mathbb{Z}[c_1, \cdots, c_k]$
2. If L is a complex line bundle over X, $c_1(L)$ under usual isomorphism $\text{Pic}\ (X) \cong H^2(X, \mathbb{Z})$
3. For arbitrary complex vector bundles E of rank k over X, we define $c(E)$, called the total Chern class of E, to be $1 + f^*(c_1) + f^*(c_2) + ...$ which has the property that $c(E) = c(F)$ if $E \cong F$, $c(E)c(F) = c(E \oplus F)$, $c(\text{trivial bundle}) = 1$

Note. These facts follow from above:

1. If E is a rank k complex vector bundle over X, we can think of E as a real vector bundle(realification) of rank $2k$, then $w_{2j}(E) = c_j(E) \text{mod}\ 2$, $w_{2j+1}(E) = 0$
2. $w_1(L_1 \otimes L_2) = w_1(L_1) + w_1(L_2)$, $c_1(L_1 \otimes L_2) = c_1(L_1) + c_1(L_2)$

Theorem 9.3 (Lerary-Hirsch theorem) $F \xrightarrow{\iota} E \xrightarrow{p} B$ is a fiber bundle, $H^*(F, R)$ is finitely generated free R-module, and $\exists c_j \in H^{k_j}(E, R)$ such that $\iota^*(c_j)$ form a basis for $H^*(F, R)$. Then $H^*(B, R) \otimes_R H^*(F, R) \to H^*(E, R)$, $\sum b_i \otimes \iota^*(c_j) \mapsto p^*(b_i) \smile c_j$ is an isomorphism

Theorem 9.4 (Splitting principle) Informal statement: For formulas involving characteristic classes, you can pretend that all vector bundles are direct sum of line bundles

Formal statement: Let X be a compact Hausdorff space, $E \to X$ a rank k vector bundle(over $\mathbb{R}$ or $\mathbb{C}$), then there is another space Y_E with a surjective projection $p : Y_E \to X$ such that $p^* : H^*(X) \to H^*(Y_E)$ is injective, and $p^*(E)$ splits as a direct sum of line bundles

Proof. Idea: Let Y_E be the fiber bundle over X whose fiber over $x \in X$ is the flag manifold of E_x which can be identified with $U(k)/U(1)^k$ or $O(k)/O(1)^k$. By Leray-Hirsch theorem, p^* is injective on cohomology(with $\mathbb{Z}$ coefficients in the complex case and $\mathbb{Z}/2$ in the real case)

We get the formula for behavior under tensor products: if $E = L_1 \oplus \cdots \oplus L_k$, $F = M_1 \oplus \cdots \oplus M_l$, then $E \otimes F \cong \oplus_{j,m} L_j \otimes M_m$, so $C(E \otimes F) = \prod_{j,m}(1 + c_1(L_j) + c_1(M_m))$, $c(E) = \prod_j (1 + c_1(L_j))$, we can express c_k using elementary symmetric functions

□

What is this good for? K theory is about stable classification of vector bundles. Sometimes this is hard to compute, but we have a calculational tool: E is stably isomorphic to F implies that E and F has the same characteristic classes

Example 9.5 Given line bundles L, L', $c_1(L) = x, c_1(L') = y$, $c(L \otimes L') = 1 + (x + y) + xy$, so if a rank 2 bundle E is to be stably isomorphic to $L \oplus L'$, we need to have $c_1(E) = x + y$ and $c_2(E) = xy$, this is necessary but not sufficient

Example 9.6 Over S^2, we have complex vector bundles of rank 1, $L_n, n \in \mathbb{Z}$ with $c_1(L_n) = nx$, x is the usual generator of $H^2(S^2)$. If we realify L_n, there are only two stable isomorphism classes, since L_n is stably isomorphic to L_{n+2}

10 Other categories of vector bundles

The subject of K-theory is useful in studying many different categories of vector bundles. A useful framework is that of a **ringed sapce** $(X, \mathcal{O}_X)$. Here X is a topological space (not necessarily Hausdorff) and $\mathcal{O}_X$ is a **sheaf of rings** on X. That means that for each open set $U \subseteq X$, we have a ring $\mathcal{O}_X(U)$ (to be thought of as the allowable functions on U) and that we have compatible **restriction maps**, in general neither injective nor surjective $\mathcal{O}_X(U) \to \mathcal{O}_X(V)$ when $V \subseteq U$. The key axioms enable one to **recover a function from its restrictions**:

1. If $\{U_i\}$ is a covering of U, then $\mathcal{O}_X(U) \to \prod_i \mathcal{O}_X(U_i)$ is injective
2. If $\{U_i\}$ is a covering of U and one is given $f_i \in \mathcal{O}_X(U_i)$ such that $f_i \mid_{U_i \cap U_j} = f_j \mid_{U_i \cap U_j}$ for all i and j, then there exists $f \in \mathcal{O}_X(U)$ restricting to f_i on U_i (gluing condition)

In all cases we will be interested in, $(X, \mathcal{O}_X)$ will be what Weibel calls **locally ringed**, where we add two more axioms:

1. $\mathcal{O}_X$ is a sheaf of **commutative rings**
2. for $x \in X$, $\mathcal{O}_{X,x} = \lim\limits_{\substack{\to \\ x \in U}} \mathcal{O}_X(U)$ is a local ring
3. the topology on X is T_0, i.e. distinct points have distinct families of open neighborhoods. (Otherwise we can pass to a quotient space)

Example 10.1

- X a topological (say compact Hausdorff) space and $\mathcal{O}_X(U) = \{\text{continuous functions } U \to \mathbb{F}\}$, where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. In this case, $\mathcal{O}_{x,x}$ is the local ring of **germs of continuous functions** at x.
- R a commutative ring. $X = \operatorname{Spec} R$, $\mathcal{O}_X(X) = R$, $\mathcal{O}_X(D(f)) = R[\frac{1}{f}]$, $\mathcal{O}_{X,\mathfrak{p}} = R_{\mathfrak{p}}$. This is a key example called an **affine scheme**.
- a locally ringed space $(X, \mathcal{O}_X)$ such that X has an open covering $\{U_i\}$ with $(U_i, \mathcal{O}_X \mid_{U_i})$ an affine scheme for each i. Such a ringed space is called a **scheme**. Aside from affine schemes themselves, the key examples are **projective schemes** over a field k. These are subschemes of **projective space** $\mathbb{P}^n_k$, which is constructed from the ring of homogeneous polynomials in $n+1$ **homogeneous coordinates** $x_0, \cdots, x_n$. Note that $\mathbb{P}^n_k$ has a covering $\{U_i\}_{i=0}^n$, where $U_i = \{x_i \neq 0\}$. Each U_i is isomorphic to **affine n-space** $\mathbb{A}^n_k = \operatorname{Spec} k[x_1, \cdots, x_n]$.
- A **stein manifold** X, i.e. a complex analytic submanifold of $\mathbb{C}^n$ for some n. These are the analytic counterparts of (smooth) affine schemes. The sheaf $\mathcal{O}_X$ is defined by $\mathcal{O}_X(U) = \{\text{holomorphic functions on } U\}$
- More generally, an **analytic space** X, with $\mathcal{O}_X(U) = \{\text{holomorphic functions on } X\}$

10.1 Vector bundles on a ringed space

Let $(X, \mathcal{O}_X)$ be a locally ringed space like one of the examples above. A **vector bundle** over X of **rank** k is a sheaf $\mathcal{F}$ of $\mathcal{O}_X$-modules (i.e., for each U open in X, $\mathcal{F}(U)$ is a module over $\mathcal{O}_X(U)$, and the sheaf axioms hold) which is **locally free** of rank k, i.e., such that there is an open covering $\{U_i\}$ with $\mathcal{F}(U_i) \cong \mathcal{O}_X(U_i)^k$. In the cases of affine schemes and Stein manifolds, a vector bundle $\mathcal{F}$ over X is equivalent to a finitely generated projective module over $R = \mathcal{O}_X(X)$. However, this fails badly over projective schemes and compact complex manifolds, since then R is much too small (it consists only of constant functions).

10.2 Classifying vector bundles

Suppose $\mathcal{F}$ is a rank k vector bundle over $(X, \mathcal{O}_X)$, choose $\{U_i\}$ with $\mathcal{F}(U_i) = \mathcal{O}_X(U_i)^k$. Over $U_{ij} = U_i \cap U_j$, we have an isomorphism $\mathcal{F}(U_i) \mid_{U_{ij}} \overset{\cong}{\to} \mathcal{F}(U_j) \mid_{U_{ij}}$, which we can identify with a regular function over U_{ij} with values in bundle automorphisms. This gives a Cech cocycle with values in $\mathrm{GL}_k(\mathcal{O}_X)$, and so we get the Classification theorem

Theorem 10.2 $\{\mathrm{VB}(X, \mathcal{O}_X)\} \cong H^1(X, \mathrm{GL}_k(\mathcal{O}_X))$. The right hand side is somewhat mysterious unless $k = 1$, when it is the cohomology of the **multiplicative group** sheaf $\underline{\mathbb{G}_m}$ or $\mathcal{O}_X^\times$. So we have Pic $(X, \mathcal{O}_X) \cong H^1(X, \mathcal{O}_X^\times)$

10.3 Chern classes of line bundles

Suppose X is an analytic manifold, a Stein manifold, a compact complex manifold, or the underlying analytic space of a projective variety over $\mathbb{C}$ (algebraic variety = reduced irreducible scheme). Let $X_{\mathrm{top}} = X$ just as a topological space, the cohomology of X is the target for Chern classes of topological vector bundles. Note that there is a natural map (forget analytic or algebraic structures) $\mathrm{VB}(X, O_X) \to \mathrm{VB}(X_{top})$. So via this forgetful map, we can define Chern classes. But the forgetful map need not be either injective nor surjective, i.e. many algebraic vector bundles can give rise to the same topological vector bundle, and some topological vector bundles can't be made algebraic.

Another useful construction: The **exponential sequence**. This works in the analytic but not in the algebraic category. Say X is a connected complex analytic space or a complex manifold, $\mathcal{O}$ is the sheaf of holomorphic functions, then we have an exact sequence of sheaves:

$$0 \to \mathbb{Z} \overset{2\pi i}{\hookrightarrow} \mathcal{O} \overset{\exp}{\to} \mathcal{O}^\times \to 1$$

This gives a long exact sequence in sheaf cohomology

$$0 \to \mathbb{Z} \to H^0(X, O) \to H^0(X, O^\times) \to H^1(X, \mathbb{Z}) \to H^1(X, \mathcal{O}) \to H^1(X, O^\times) = \mathrm{Pic}\ (X, O) \overset{c_1}{\to} H^2(X, \mathbb{Z})$$

There are cases where $c_1 : \mathrm{Pic}\ (X, O) \to H^2(X, \mathbb{Z})$ is an isomorphism, then the classification of analytic and topological line bundles is the same

Example 10.3 $X = \mathbb{P}^1_{\mathbb{C}}$, every analytic vector bundle is determined by its Chern class, and all Chern classes can be realized. So the analytic line bundles are all of the form $\mathcal{O}(n)$, $n \in \mathbb{Z}$

Theorem 10.4 (Highly non-trivial, GAGA) For complex projective varieties, classifications of algebraic and analytic vector bundles coincide

10.4 More on Algebraic and holomorphic vector bundles

Example 10.5 On $X = \mathbb{P}^1_k$, Pic $(\mathbb{P}^1_k) \cong \mathbb{Z}$ with line budnles denoted by $O(m)$, $m \in \mathbb{Z}$, whose local sections are homogeneous Laurent polynomials of degree m. If $k = \mathbb{C}$, use exponential sequence we have

$$0 \to H^1(\mathbb{P}^n_{\mathbb{C}}, \mathcal{O}_X) \to H^1(\mathbb{P}^n_{\mathbb{C}}, \mathcal{O}_X^\times) = \mathrm{Pic}\ (\mathbb{P}^n_{\mathbb{C}}) \overset{c_1}{\to} H^2(\mathbb{P}^n_{\mathbb{C}}, \mathbb{Z}) \to H^2(\mathbb{P}^n_{\mathbb{C}}, \mathcal{O}_X)$$

The higher cohomology $H^j(\mathbb{P}^n_k, \mathcal{O}_X)$ vanishes for $j > 0$, so the Chern class map c_1 is an isomorphism

10.5 Similarities and differences between algebraic and topological categories

1. The same topological bundles may admit many inequivalent algebraic structures. Or it may admit none at all.
2. Exact sequences of algebraic vector bundles **do not split** in general, this is different from the projective modules over a ring.

Example 10.6

1. E Elliptic curve, say over $\mathbb{C}$. We have the exponential exact sequence

$$0 \to \mathbb{Z} \overset{2\pi i}{\to} \mathbb{C} \overset{\exp}{\to} \mathbb{C}^\times \overset{0}{\to} H^1(E,\mathbb{Z}) \to H^1(E,\mathcal{O}) \to \text{Pic } (E) \to H^2(E,\mathbb{Z}) \to H^2(E,\mathcal{O})$$

   Here E is topologically $S^1 \times S^1$, so $H^1(E,\mathbb{Z}) = \mathbb{Z}^2$, $H^2(E,\mathbb{Z}) = \mathbb{Z}$, we have $H^1(E,\mathcal{O}) = \mathbb{C}$, $H^2(E,\mathcal{O}) = 0$. So we get the short exact sequence

$$0 \longrightarrow \mathbb{C}/\mathbb{Z} \times \mathbb{Z} \longrightarrow \text{Pic } (E) \underset{\longleftarrow}{\longrightarrow} \mathbb{Z} \longrightarrow 0$$

   and Pic $(E) \cong \mathbb{T}^2 \times \mathbb{Z}$. In particular, the Chern class c_1 does not determine the class of a line bundle.

2. $X = P^1_k$, every vector bundle over X splits as a direct sum of line bundles. But unlike the case of complex top bundles over $S^2 = P^1_{\mathbb{C}}$, not every rank two vector bundle splits as $V = \det(V) \oplus \text{trivial}$. For example, $\mathcal{O} \oplus \mathcal{O}$ and $\mathcal{O}(1) \oplus \mathcal{O}(-1)$ are both topologically trivial, but they are algebraically distinct.

3. Even on $\mathbb{P}^1$, we have exact sequence

$$0 \to \mathcal{O}(-2) \to \overset{\alpha}{\to} \mathcal{O}(-1) \oplus \mathcal{O}(-1) \overset{\beta}{\to} \mathcal{O} \to 0$$

   doesn't split. Since there are no morphisms from $\mathcal{O}(p) \to \mathcal{O}(q)$, $p > q$. Here $\beta(f,g) = x_0 f + x_1 g$, $\alpha(h) = (x_1 h, -x_0 h)$

10.6 Invertible ideal sheaves and divisors

Let X be an integral scheme, i.e., one such that $\mathcal{O}_X(U)$ is an integral domain for each affine open set U. In particular, X is connected and reduced, there is a common field of fractions for all $\mathcal{O}_X(U)$, called $k(X)$, the field of rational functions

Example 10.7 $X = \mathbb{P}^n_k$, $k(X) = k(x_0, \cdots, x_n)$. $\mathcal{K}$ is the constant sheaf $k(X)$. A **fractional ideal sheaf** $\mathcal{I}$ is a $\mathcal{O}$-submodule of $\mathcal{K}$ containded in $f\mathcal{O}$ for some $f \in k(X)$. It's called invertible if $\mathcal{I}\mathcal{J} = \mathcal{O}$ for some other fractional ideal sheaf $\mathcal{J}$. Then it's a line bundle (invertible locally free sheaf) with dual line bundle $\mathcal{J}$

Proposition 10.8 X integral scheme, then there is an exact sequence

$$1 \to H^0(X, \mathcal{O}^\times) \to k(X)^\times \to \text{Cart } (X) \to \text{Pic } (X) \to 1$$

with Cart (X) the group of invertible ideal sheaves. (Note that Pic $(X) =$ Cart $(X)/\sim$)

In case X is integral, separated, and locally factorial (local rings are all UFDs), there is an isomorphism between Cartier and Weil divisors (integral linear com of closed integral subschemes of codimension 1)

11 Weibel chapter II on K_0

M is an abelian monoid, F is the forgetful functor, the group completion functor G is the left adjoint of F, i.e. Hom $(M, F(A)) \cong$ Hom $(G(M), A)$. If M is a semiring, then $G(M)$ would be a ring. $G(M)$ consists of formal expressions $[m] - [m']$, with $([m] - [m']) + ([n] - [n']) = [m+n] - [m+n]'$, $([m] - [m']) \otimes ([n] - [n']) = [m \otimes n] - [m \otimes n'] - [m' \otimes n] + [m' \otimes n']$, and $[m] - [m'] = [n] - [n'] \Leftrightarrow m + n' + p = n + m + p$ for some $p \in M$

Example 11.1 X compact Hausdorff, $M = \text{VB}(X)$, either over $\mathbb{R}$ or $\mathbb{C}$, $G(M) = K^0(X)$

Example 11.2

1. $M = \mathbb{N}$, $G(M) = \mathbb{Z}$.
2. R is an integral domain, $M = R - \{0\}$, $G(M) = \text{Frac } (R)^\times$
3. X is a compact Hausdorff space, $M = \text{VB}(X)$, the isomorphism classes of vector bundles over X (say over either $\mathbb{R}$ or $\mathbb{C}$), $G(M)$ is called $K^0(X)$
4. X scheme, $M = \text{VB}(X)$, $G(M)$ is called $K^0(X)$ (may differ from the topological case, even with the same notation). When X is a projective variety over $\mathbb{C}$, we have a forgetful map $K^0(X) \to K^0(X_{\text{top}})$
5. R is a commutative ring, M is the semiring of isomorphism classes of finitely generated projective R-modules. The multiplication comes from $\otimes_R$, Then $K_0(R) = G(M)$ is a commutative ring. The product operation is called the **cup product**. For example, if $R = C^{\mathbb{R}}(S^2)$, we saw that M is graded by rank and $M_n \cong \begin{cases} \mathbb{Z}, n=2 \\ \mathbb{Z}/2, n \geq 3 \end{cases}$. So $K_0(R) \cong \mathbb{Z} \oplus \mathbb{Z}/2$ as an abelian group, with the isomorphism given by the rank and the Stiefel-Whitney class w_2. Taking tensor product with a (stably) trivial bundle doesn't change w_2. On the other hand, if $w_2(E) \neq 0$ and $w_2(F) \neq 0$, then $w_2(E \oplus F) = w_2(E) + w_2(F) = 0$ and $w_2(E \otimes F) = 0$. So as a ring $K_0(R) = \mathbb{Z}[t]/(2t, t^2)$
6. G finite group, M is the commutative semiring(under $\oplus, \otimes$) of isomorphism classes of finite dimensional representations of G, the group completion is the representation ring $R(G)$ which is a free abelian group with rank the number of conjugacy classes. $R(G) \to \mathbb{Z} = R(\{1\})$ by restriction is the augmentation map, kernel being the augmentation ideal. Representation ring plays an important role in equivariant K-theory. If $G = \mathbb{Z}/k$, all irreducible representations are characters of $G^\vee$, $R(G) = \mathbb{Z}[t]/(t^k - 1)$, t maps the generator of G to $e^{\frac{2\pi i}{k}}$
7. G finite group, M is the commutative semiring(under $\sqcup, \times$) of finite G sets, the group completion is $A(G)$, called the Burnside ring, which free abelian group with rank the number of conjugacy classes of subgroups $H \leq G$, there is a natural ring map $A(G) \to R(G)$, sending a G-set X to the representation of G on $\mathbb{C}X$

11.1 Basics of $K_0(R)$

R is a ring, not necessarily commutative, $K_0(R)$ is the Grothendieck group of the monoid of isomorphism classes of finitely genearated projective R-modules. We can't drop the finitely generated condition because of the **Eilenberg swindle**: Suppose P is a projective module over R, with $P \oplus Q$ a free module. Claim that P is always stably free(if we don't restrict to finite rank). Since free module $(P \oplus Q) \oplus (P \oplus Q) \oplus \cdots$ can be written as $P \oplus (Q \oplus P) \oplus (Q \oplus P) \oplus \cdots = P \oplus$ free module. Not only that, but sometimes for some rings, $K_0(R) = 0$

Definition 11.3 (Karoubi) A ring R is called **flasque**(French word meaning flabby) if there is an R-bimodule M, finitely generated and projective as a right R-module, such that $R \oplus M \cong M$ as R-bimodules

Proposition 11.4 (Karoubi) If R is flasque, then $K_0(R) = 0$. In fact, all of its K groups vanishes

Proof. Let P be a finitely generated right R-module, then $P \oplus (P \otimes_R M) = P \otimes_R (R \oplus M) \cong P \otimes_R M$, so P is stably isomorphic to 0

□

Example 11.5 End (R^∞) is flasque, with M being itself

11.2 Von Neumann regular rings

A ring R (not necessarlily commutative) is called **Von Neumann regular** (the notion was introduced by Von Neumannm PNAS, 1936) if $\forall a \in R$, $\exists x \in R$ with $axa = a$, x behaves like an inverse for a. Any division ring D is von neumann regular ($x = 0$ if $a = 0$, and $x = a^{-1}$ if $a \neq 0$), and $M_n(D)$ is also Von Neumann regular

Proposition 11.6 If R is Von Neumann regular, then $\lambda : I \to$ left annihilator of I, $\rho : J \to$ right annihilator of J. This gives order reversing maps between left and right ideals

Proof. Obviously $I_1 \subseteq I_2 \Rightarrow \lambda(I_2) \subseteq \lambda(I_1)$, and similarly for ρ. $I \subseteq \rho\lambda(I)$, $J \subseteq \lambda\rho(J)$. So $\lambda\rho\lambda(I) \subseteq \lambda(I)$, but on the other hand $\lambda(I) \subseteq \lambda\rho\lambda(I)$, so $\lambda = \lambda\rho\lambda$, similarly $\rho = \rho\lambda\rho$

□

Proposition 11.7 If R is Von Neumann regular, then each principal left ideal $Ra = Re$ for some idempotent e, and the right annihilator of Ra is $(1-e)R$. Similarly, any $aR = eR$ for some idempotent e, and the left annihilator of aR is $R(1-e)$. Thus there is an order revering bijection between left and right principal ideals.

Proof. choose x such that $axa = a$, then $e = xa$, $Ra = Re$. Re and $(1-e)R$ are each other's annihilators

□

Proposition 11.8 Suppose R is Von Neumann regular, then there are some orthogonal idempotentes e, f such that $aR = eR$, $bR = fR$ and $aR + bR = eR + fR = (e+f)R$

Proof. By Proposition 11.7, there exists an idempotent e such that $aR + bR = eR + bR = eR + (1-e)bR$, choose f_1 such that $(1-e)bR = f_1R$, clearly $ef_1 = 0$, let $f = f_1(1-e)$, then $ff_1 = f_1$, so $f^2 = f$. $aR + bR = eR + f_1R = eR + fR$

□

Corollary 11.9 If R is Von Neumann regular, then every finitely generated right projective R-module is a direct sum of eR's with e being idempotents, and in particular every finitely generated right ideal is projective

It is often easy to compute K_0 in this case

11.3 More about K_0 of rings

If $R = \lim_{\to} R_i$, then $K_0 = \lim_{\to} K_0(R_i)$

Example 11.10 An AF or AFD(approximately finite dimensional) rings is a ring of the form $\lim_{\to} R_i$, where R_i is a finite dimensional semisimple algebra over $\mathbb{C}$. i.e. a direct sum of matrix algebras. Note that $K_0(R_i) = \mathbb{Z}^{n_i}$, where n_i is the number of matrix summands. maps between such algebras can be described by *Bratteli diagrams*

Thus any countable torsion-free abelian group can arise as K_0 of an AF algebra examples: the CAR algebra(canonical anticommutation relations) 1–2–4–8–..., K_0 is $\mathbb{Z}[1/2]$ and gauge invariant CAR algebra(or Pascal's triangle algebra), K_0 is countable torsion free abelian but not finitely generated

Lemma 11.11 (Idempotent lifting) R ring, I either nilpotent or complete ideal($\sum_{n=1}^{\infty} x_n$ with $x_n \in I^n$ converges to a unique element of I). Then any idempotent $\dot{e} \in R/I$ can be lifted to an idempotent e in R. Furthermore, any two such lifts are conjugate under some $u \in 1 + I$

Proof. Choose $e_1 \in R$ lifting $\dot{e}$, then $e_1^2 - e_1 \in I$. take $e = ue_1^k$

If instedad, I is complete, we replacfe $e_1 byue_1$, where $u = (e_1^2 + f_1^2)^{-1}$, which commutes with e_1 and f_1. Now $u(e_1^2 + f_1)^2 = 1$. $(ue_1)^2 + (uf_1)^2 = u$

□

Lemma 11.12 R is a ring, $\dot{u} \in R/I$ which is invertible. Then $\begin{bmatrix} \dot{u} & \\ & \dot{u}^{-1} \end{bmatrix}$ lift to something in $\mathrm{GL}_2(R)$

Proof. $\begin{bmatrix} \dot{u} & \\ & \dot{u}^{-1} \end{bmatrix} = \begin{bmatrix} 1 & \dot{u} \\ & 1 \end{bmatrix} \begin{bmatrix} 1 & \\ -\dot{u}^{-1} & 1 \end{bmatrix} \begin{bmatrix} 1 & \dot{u} \\ & 1 \end{bmatrix} \begin{bmatrix} & -1 \\ 1 & \end{bmatrix}$

□

Theorem 11.13 R is a ring with ideal I which is either nilpotent or complete. Then $K_0(R) \cong K_0(R/I)$, via the natural map $R \to R/I$

Proof. Surjectivity: Let P be a finitely generated projective (R/I)-module, then P comes from an $n \times n$ idempotent matrix $\dot{e}$ over R/I. Now $\dot{e} \in M_n(R/I) \cong M_n(R)/M_n(I)$ and note that $M_n(I)$ is either nilpotent(if I is) or complete (if I is). This requires checking, but if $a \in M_n(I)$, then powers of a have entries in powers of I, apoly Lemma 11.11 to $M_n(R), M_n(I)$ shows P comes from a finitely generated R-module
Injectivity: Suppose e, e' are idempotents in $M_n(R)$ which map to conjugate idempotents in $M_n(R/I)$. Replacing e' by a conjugate, can assume $e = e' \bmod I$(Note: if $\dot{e}$ is conjugate to $\dot{e}'$ under $\dot{u} \in \mathrm{GL}_n(R/I)$, we can't necessarily lift u to an invertible matrix in $M_n(R)$). But you can lift $\begin{bmatrix} \dot{u} & \\ & \dot{u}^{-1} \end{bmatrix} \in \mathrm{GL}_{2n}(R/I)$ to something invertible element in $\mathrm{GL}_2(R)$. So at the expense of increasing n, you can achieve $e - e' \in M_n(I)$. Then use the Lemma to show e, e' conjugate in $M_n(R)$

□

11.4 The K_0 exact sequence and Exision for K_0

We now want to show that K_0 behaves like the begining of a "homology theory" for rings. Axioms for homology of spaces:

1. Long exact sequences
2. Functoriality
3. Homotopy invariance
4. Normalization
5. (depends on what homology theory you use) behavior under limits
6. Excision

It turns out that K_0 satisfies all of these (in a suitable sense) except for homotopy invariance, even that one doesn't fail by much. Now we want to explain these. Some (normalization, behavior under limits) are fairly easy. Usual homology is normalized by requiring $H_0(\text{point}) = \mathbb{Z}$, $H_j(\text{point}) = 0$ for $j \neq 0$. We will eventually have a whole sequence K_j of functors from rings to abelian groups and the second part of this (the values of $K_j(\mathbb{Z})$, $j \neq 0$, $\mathbb{Z}$ being the "simplest" ring(Since all rings come with a canonical ring map $\mathbb{Z} \to R$)) will fail. In fact, the $K_j(\mathbb{Z})$ for $j > 1$ are quite difficult to compute and tied up with a lot of deep number theory. But $K_0(\mathbb{Z})$ is as we expect, in fact we have

Proposition 11.14 If R is a PID, then $K_0(R) \cong \mathbb{Z}$, the isomorphism being the rank

Proof. Any finitely generated torsion-free module over R (and certainly any finitely generated projective module) is free and characterized by its rank.

□

Functoriality is clear. If $\varphi : R \to S$ is a ring homomorphism sending 1 to 1, then we have an induced map $\varphi_* : K_0(R) \to K_0(S)$, and in this way K_0 is a functor from rings to abelian groups. The point here is that if P is a finitely generated projective R-module, say $P \oplus Q \cong R^n$, then $(S \otimes_\varphi P) \oplus (S \otimes_\varphi Q) \cong S \otimes_\varphi R^n = S^n$, so $\varphi_*(P) = S \otimes_\varphi P$ is finitely generated projective, and

we get functoriality from associativity of the tensor product. In fact we even have functoriality under **non-unital** ring homomorphisms. Suppose $\varphi : R \to S$ is a non-unital ring homomorphism. Then $\varphi(1_R) = e$ is an idempotent in S, and φ sends the free R-module to the projective (not necessarily free) S-module defined by e. Similarly, if P is a direct summand in R^n, then $S \otimes_\varphi P$ is a direct summand in $(Se)^n \subseteq S^n$, so again $\varphi_*(P)$ is finitely generated projective.

Let R be a aring with dieal I. Define a new ring (**double**) $D = \{(x, y) \in R \times R, x - y \in I\}$. Projection p_1 onto the first coordinate gives a split surjection of rings

$$D \underset{p_1}{\overset{\Delta}{\rightleftarrows}} R$$

, where $\Delta(x) = (x, x)$. So as R-modules, $D \cong R \oplus I$. Define **relative K_0** by $K_0(R, I) = \ker(K_0(D) \overset{(p_1)_*}{\to} K_0(R))$.

Theorem 11.15 There is an exact sequence

$$\text{GL }(R) \to \text{GL }(R/I) \overset{\partial}{\to} K_0(R, I) \overset{(p_2)_*}{\to} K_0(R) \to K_0(R/I)$$

"Excision": $\phi : R \to S$, $I \cong \phi(I)$, then $\phi_* : K_0(R, I) \to K_0(S, \phi(I))$ is an iso We can define $K_0(I)$ to be $K_0(R, I)$ for any ring R containing I as an ideal. R could be $I_+ = I \oplus \mathbb{Z}$ with $(i, n) \cdot (j, m) = (ij + nj + mi, nm)$

Proof.

□

What about **homotopy invariance**?

The analogue of the homotopy invariance in algebra would be invariance under taking polynomials (This is called $\mathbb{A}^1$ homotopy invariance in algebraic geometry, since $\mathbb{A}^1_k = \text{Spec } k[t]$ is the algebraic geometry analogue of $[0, 1]$, and homotopy invariance says multipliying with $[0, 1]$ doesn't change anything). What happens is that if R is a ring, we get a split injection

$$K_0(R) \underset{t \mapsto 0}{\overset{}{\rightleftarrows}} K_0(R[t])$$

. In general, this is not an isomorphism, and $K_0(R[t]) \cong K_0(R) \oplus NK_0(R)$. However, if R is nice enough (e.g. Spec R a smooth scheme), then $NK_0(R) = 0$

12 Traces and Dimensions

While we've defined $K_0(R)$ as an abstract group, sometimes one would like **numerical invariants** that enable one to detect or even identify elements of this group. For example, if one has two vector bundles over a space (or ringed space like Spec R), are there numerical invariants that enable one to determine when they are stably isomorphic? Here are some examples:

1. R is a PID, rank : $K_0(R) \overset{\cong}{\to} \mathbb{Z}$
2. (much less trivial) $R = C^{\mathbb{C}}(S^2)$, $K_0(R) \underset{(\text{rank}, w_2)}{\overset{\cong}{\to}} \mathbb{Z} \oplus \mathbb{Z}$, or as a ring $\mathbb{Z}[t]/(t^2)$
3. (still less trivial) $R = C^{\mathbb{R}}(S^2)$, $K_0(R) \underset{(\text{rank}, w_2)}{\overset{\cong}{\to}} \mathbb{Z} \oplus \mathbb{Z}/2$, or as a ring $\mathbb{Z}[t]/(2t, t^2)$
4. R is a Boolean ring, so Spec R is totally disconnected, $K_0(R) \overset{\cong}{\to} C(\text{Spec } R, \mathbb{Z})$

Let's start with a more general notion of dimension that will encompass (4).

Definition 12.1 Let R be a commutative ring, $H_0(R) = C(\text{Spec } R, \mathbb{Z})$. This is a free abelian group.

Proposition 12.2 Let R be a commutative ring, then the **rank** or **dimension** function $K_0(R) \to H_0(R)$ is split surjective. The kernel of the map is called **reduced K-theory**, $\tilde{K}_0(R)$. If R is an integral domain, or if R has no non-trivial idempotents (this is automatic if $R_{\text{red}} = R/$ nilrad (R)) is an integral domain, then $H_0(R) = \mathbb{Z}$

Example 12.3 (Topological reduced K-theory) Let X be a connected compact Hausdorff space, $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, $R = C^{\mathbb{F}}(X)$. Then $H_0(R) = \mathbb{Z}$ since R has no non-trivial idempotents (even though it's very far from being an integral domain).

In this case, recall that isomorphism classes of rank k vector bundles over X are classified by $[X, BU(k)]$ or $[X, BO(k)]$, homotopy classes of maps from X to the infinite Grassmannian $\lim_{\to} \mathrm{Gr}_{\mathbb{F}}(k, n)$. In this case, $\tilde{K}_0(R)$, written also $tilde K_{\mathbb{F}}(X)$, is the group of stable isomorphism classes of virtual vector bundles of (virtual) rank 0. Since every such class can be written as $[E] - [1_k]$, where E is a rank k bundle and 1_k is the trivial rank k bundle, we see that $\tilde{K}_{\mathbb{C}}(X) = [X, BU(k)]$ and $\tilde{K}_{\mathbb{R}}(X) = [X, BO(k)]$

Aside from looking just at "rank", sometimes we can get more subtle information from "trace". For any ring R (not necessarily commutative - the noncommutative case is much more interesting), define the **Hattori-Stallings trace map** as follows. Let $[R, R]$ be the additive subgroup of R generated by commutators $rs - sr$. The trace of a matrix defines an isomorphism $M_n(R)/[M_n(R), M_n(R)] \xrightarrow{\cong} R/[R, R]$. Note that these are just additive groups, the multiplicative structure has been lost. Now one gets

$$\mathrm{tr} : K_0(R) \to R/[R, R]$$

by sending $[R^n e]$ (e an idempotent in $M_n(R)$), to tr (e) modulo commutators

Proposition 12.4

13 Lambda rings and Adams operations

Another piece of structure on $K_0(R)$ when R is a commutative ring comes from the operation $E \mapsto \bigwedge^j(E)$ on vector bundles. When $j = \text{rank } E$, this is what we previously called the **determinant**

Definition 13.1 A is a **λ-ring**(for Grothendieck, a **pre-λ-ring**) if there are maps $\lambda^j : A \to A$, $j \geq 0$, satisfying

1. $\lambda^0(x) = 1$
2. $\lambda^1(x) = x$
3. $\lambda^k(x + y) = \sum_{i+j=k} \lambda^i(x)\lambda^j(y)$

Example 13.2 $A = \mathbb{Z}$, $\lambda^j(n) = \binom{n}{j}$. It follows from the identities

$$\left(\sum_{j=0}^{\infty} \binom{n}{j} t^j\right)\left(\sum_{k=0}^{\infty} \binom{m}{k} t^k\right) = (t+1)^{n+m} = \sum_{j=0}^{\infty} \binom{m+n}{j} t^j$$

Example 13.3 R is a commutative ring, $A = K_0(R)$ which is the group completion of Proj (R) (the semiring of isomorphism classes of finitely generated projective R-modules), define $\lambda^j[P] = [\wedge^j(P)]$. Note $\lambda^0[P] = [\bigwedge^0 P] = [R]$, $\lambda^1[P] = [\bigwedge^1 P] = [P]$, and $\lambda^k[P \oplus Q] = \sum_{i+j=n} \lambda^i[P]\lambda^j[Q]$ follows from the decomposition $\bigwedge^k(P \oplus Q) = \oplus_{i+j=n} \bigwedge^i(P) \bigwedge^j(Q)$. These operations extend to the group completion A because $[P] \mapsto \sum_j \lambda^j[P]t^j$ is a monoid homomorphism from Proj (R) to $A[[t]]$, since

$$\left(\sum_{j=0}^{\infty} \lambda^j[P]t^j\right)\left(\sum_{k=0}^{\infty} \lambda^k[P]t^k\right) = \sum_{n=0}^{\infty} \sum_{i+j=n} \lambda^i[P]\lambda^j[Q]t^n = \sum_{n=0}^{\infty} \lambda^n[P \oplus Q]t^n$$

Definition 13.4 I is **λ-ideal** if $\lambda^j(I) \subseteq I$ for all $j \geq 1$. (Recall $\lambda^0 : A \to \{1\}$, so we can't expect this for $j = 0$). A **binomial ring** is a commutative ring A contained in a $\mathbb{Q}$-algebra $A_{\mathbb{Q}}$ with property that if $k \geq 1$, $x \in A$, then $\binom{x}{k}$ lies in A and not just in $A_{\mathbb{Q}}$. Example $A = \mathbb{Z}$

Definition 13.5 A be a λ-ring. We say it has a **positive structure** if

1. there is a λ-subring H^0 which is a binomial ring
2. there is a surjective λ ring homomorphism $\epsilon : A \to H^0$ which is identity on H^0, called augmentation
3. there is a λ-subsemiring P of A containing $\mathbb{N}$, such that every element in $\tilde{A} = \ker \epsilon$ can be written as $p - q$ for $p, q \in P$
4. if $p \in P$, then $\epsilon(P) \in \mathbb{N}$, and $\lambda^i(p) = 0$ for $i > \epsilon(P)$, furthermore, $\lambda^{\epsilon(P)}(P)$ (the "determinant" of p) is a unit in A

When we have prositive structure, the **line elements** (analogues of line bundles) of A are elements $\ell \in P$ with $\epsilon(\ell) = 1$. These are a subgroup of $A^\times$. Let L be the set of line elements

Example 13.6 If R is a commutative ring, $A = K_0(R)$ is a λ-ring with a positive structure, where $P = \text{Proj } (R)$, $H^0(R) = [\text{Spec } R \to Z]$, ϵ is the dimension function, $L = \text{Pic } (R)$

Example 13.7 G is a finite group, $A = R(G)$ is a λ-ring with positive structure. P is the isomorphism classes of finite dimensional representations of G (This is a semiring with cancellation with respect to addition, and P generates $R(G)$). $\lambda^j(\pi) = \wedge^j(\pi)$, ϵ is the dimension function(by evaluating the character at 1). The group of line elements consists of 1-dimensional representations, i.e. homomorphisms $G \to \mathbb{T}$, so it can be identified with the Pontryagin dual group to G_{ab}

13.1 Splitting principle

Definition 13.8 Let A be a λ-ring with positive structure (P, ϵ). We say that A **satisfies the splitting principle** if given $p \in P$, there is an injection $(A, P) \hookrightarrow (\tilde{A}, \tilde{P})$ of λ-rings such that p is a sum of line elements in $\tilde{P}$

Example 13.9 (Topological splitting principle, Karoubi's book on topological K theory) X is a compact Hausdorff space, $A = K_{\mathbb{F}}(X)$, P is the set of vector bundles. This satisfies the splitting principle since if $E \xrightarrow{p} X$ is a vector bundle, Y is the associated flag variety bundle over X (i.e. $Y \to X$ is a fiber bundle with Y_x = flag variety of E_x). Then $K(X) \hookrightarrow K(Y)$ and the pullback of E to Y splits as a direct sum of line bundles

In the topological setting, the advantage of the splitting principle is that it made it easy to define Chern classes. In the λ-ring setting, it gives us a formula for λ^j using the fact that for $\ell_1, \cdots, \ell_n$ line elements, $\lambda^j(\ell_1 + \cdots + \ell_n) = \sum_{k_1 < \cdots < k_j} \ell_{k_1} \cdots \ell_{k_j}$

Example 13.10 (Splitting principle for $R(G)$, G is a finite group) Any element of $R(G)$ is determined by its character. To determine the character, it's enough to compute all restrictions $R(G) \to R(H)$ where H is a cyclic subgroup of G. In $R(H)$, every representation is a direct sum of 1-dimensional representations

13.2 Preview of Adams operations

Out of the λ-operations, Adams constructed the **Adams operations** ψ^j on topological K-theory, and these enable you to recover a good notion of dimension. He used these for his famous "vector fields on spheres" calculation, to compute the maximal number of linear independent sections on a generic vector bundle over a sphere, and on the tangent bundle TS^n

Theorem 13.11 Let A be a λ-ring with augmentation ϵ. (We assume $H^0 \subseteq A$ is a binomial ring, e.g. $\mathbb{Z}$, $\epsilon : A \to H^0$ is a λ-ring homomorphism which is the identity on H^0, and we have a generating subsemiring $P \subseteq A$ of **positive elements** such that $\epsilon : P \to \mathbb{N}$ and $\lambda^j(p) = 0$ for $j > \epsilon(p)$ if $p \in P$. Also, every element of $\tilde{A} = \ker \epsilon$ is a difference of two elements of P with the same ϵ-value and $\lambda^{\epsilon(p)}p$ is a unit for $p \in P$. Then there are unique **Adams operations** $\psi^k : A \to A$, $k \in \mathbb{N}$, with the following properties:

1. $\psi^0 = \epsilon$
2. $\psi^1 = \mathrm{id}$
3. $\psi^k(\ell) = \ell^k$ for ℓ a line element
4. $\psi^k(x) =$ coefficient of t^k in the formal power series $\epsilon(x) - t\frac{d}{dt}\log\lambda_{-t}(x)$, where $\lambda_t(x) = \sum_k \lambda^k(x)t^k$ and of course $\lambda_{-t}(x) = \sum_k (-1)^k\lambda^k(x)t^k$

Proof.

□

Proposition 13.12 If the λ-ring A satisfies the splitting principle, then each ψ^k is a ring homomorphism and $\psi^l\psi^k = \psi^{l+k}$

Proof.

□

Proposition 13.13 Assume the hypotheses of the previous proposition, and also that $\tilde{A} = \ker(\epsilon : A \to H^0)$ is a nilpotent ideal. (This is the case if $A = K_0(X)$ for X a finite CW complex, or if $A = K_0(R)$ for R an algebra of finite type over a field. So the hypothesis covers many examples of interest.) Then the eigenvalues of ψ^k on $A \otimes \mathbb{Q}$ are powers of k

Proof. So get a finite filtration of $A_{\mathbb{Q}}$ with $\psi^k = k^n$ on n-th graded piece, this proves the result

□

Lambda operations and Adams operations play a big role in K-theory. In particular, ψ^p for p prime behaves very much like the Frobenius automorphism in Galois theory since ψ^p is a ring homomorphism with $\psi^p(l) = \ell^p$ for line element ℓ

14 K_0 in more general contexts

Aside from K_0 of rings, (using projective modules), we will want to study things like K_0 of algebraic vector bundles, which don't always split as direct sums. For this, we want to introduce K-theory not just of rings but of suitable categories, "like" categories of vector bundles. What does "like" mean? There are several possible answers, depending on the level of abstraction you are willing to tolerate

1. Symmetric monoidal categories(Weibel II.5)
2. Abelian categories(Weibel II.6)
3. Exact categories(Weibel II.7)
4. Waldhausen categories(Weibel II.9)

We will concentrate on Abelian and exact categories, which cover most of the applications you need, at least at first.

Symmetric monoidal categories are categories $\mathcal{S}$ with $\boxtimes : \mathcal{S} \times \mathcal{S} \to \mathcal{S}$, a distringuished object e, and natural equivalences $e \boxtimes s \cong s \boxtimes e \cong s$, $(s \boxtimes t) \boxtimes u \cong s \boxtimes (t \boxtimes u)$, $s \boxtimes t \cong t \boxtimes s$. There are some compatibility axioms

Example 14.1 R is a ring, S is the category of finitely generated projective R-modules, and $\circ = \oplus$

From a symmetric monoidal category, one can construct a K_0 group. This covers K-theory of rings but also other examples

14.1 K_0 of abelian categories and G-theory (Weibel II.6)

Recall that an **additive category** $\mathcal{C}$ is a category in which Hom sets are abelian groups, composition is bilinear, and there is a distinguished initial and termial object 0 with $\text{Hom}\ (0, A) = \text{Hom}\ (A, 0) = 0$ for all $A \in ob\mathcal{C}$. In addition, we require finite products and coproducts, which coincide and are denoted $\oplus$ (There may or may not be inifinite products and coproducts, when these exists, they are usually not the same and we denote products by $\prod$)

An **abelian category** is a bit stronger, it's an additive category $\mathcal{C}$ in which **every morphism has a kernel and a cokernel**, every monic is a kernel, and every eop is a cokernel. These axioms insure that exact sequence exist and have the usual properties. The **standard example** of an abelian category is the category of left (or) right R-modules. This example is also universal

Theorem 14.2 (Freyd and Mitchell) Every abelian category embeds in a category of modules over a ring

14.2 K_0 of abelian category

$\mathcal{C}$ is a skeletally small abelian category, we define $K_0(\mathcal{C})$ be the universal abelian group on the objects (in the skeleton) of $\mathcal{C}$, subject to the axioms: $[A'] + [A''] = [A]$ if $0 \to A' \to A \to A'' \to 0$ is exact ($[0] = 0$ follows). Note that if $A' \to A$ is an isomorphism, then $[A] = [A']$

If $\mathcal{C}$ and $\mathcal{C}'$ are abelian categories and $F : \mathcal{C} \to \mathcal{C}'$ is an **exact functor**, then F induces a homomorphism $F_* : K_0(\mathcal{C}) \to K_0(\mathcal{C}')$

Example 14.3 (Main example) R is a right Noetherian ring, and $\mathcal{C}$ is the category of finitely generated right R-modules. This is an abelian category, since a submodule or quotient of a finitely generated module is finitely generated. Define $G_0(R)$ (also writen as $K_0'(R)$) to be $K_0(\mathcal{C})$

14.3 The Cartan map

The **Cartan homomorphism** is the natural map $K_0(R) \to G_0(R)$ sending $[P] \in K_0(R)$, P a finitely generated projective R-module, to the class of P in $G_0(R)$

Proposition 14.4 If R is a PID, then the Cartan map $\mathbb{Z} \cong K_0(R) \to G_0(R)$ is an isomorphism

Proof. M is a finitely generated R-module, $0 \to F' \to F \to M \to 0$ is a presentation with F, F' free, then $[M] = [F] - [F']$ lies in the image of the Cartan map. So $G_0(R)$ is a quotient of $K_0(R)$ under the Cartan map. But the map is injective since $[M] \mapsto \text{rank}\ M = \dim_F F \otimes_R M$, $F = \text{Frac}\ (R)$, gives a splitting $G_0(R) \to K_0(R)$

□

14.4 Functoriality of G_0

$f : R \to S$ is a ring homomorphism of right Noetherian rings

1. If S is flat over R, then $f_* = - \otimes_R S : G_0(R) \to G_0(S)$ is exact. Flatness is very restrictive, so we will weaken this later to the assumtion that S has a finite resolution by finitely generated projective R-modules, which is often the case in examples of interest.
2. If S is a finitely generated R-module, then we get a transfer homomorphism $f^* : G_0(S) \to G_0(R)$ by thinking of finitely generated S-modules as finitely generated R-modules

K_0 has more functoriality but harder to compute, G_0 has less but easier to compute

14.5 Examples from algebraic geometry

Lots of interesting examples of this machinery come from algebraic geometry. If $(X, \mathcal{O}_X)$ is a Noetherian scheme, then the coherent sheaves $\mathcal{F}$ over X form an abelian category Coh (X). (Recall that an $\mathcal{O}_X$-module is coherent if locally there is an exact sequence $\mathcal{O}^m \to \mathcal{O}^n \to \mathcal{F}$ for some finite m, n). Then $K_0(\text{Coh } (X))$ is denoted $G_0(X)$. It behaves like a **homology** theory for schemes, whereas $K_0(X)$ behaves like a **cohomology** theory. The Cartan map $K_0(X) \to G_0(X)$ is thus related to **Poincare duality**.

If $X = \text{Spec } R$, R Noetherian commutative ring, then Coh (X) is just the category of finitely generated R-modules, and we get back $G_0(R)$

Example 14.5 $R = \mathbb{Z}, S = \mathbb{Z}/p^n$, then S is finitely generated as an R-module, but the map $R \to S$ is definitely not flat, we know that $\mathbb{Z} \cong K_0(R) \to G_0(R)$ is an isomorphism since R is a PID, we also know that the map $\mathbb{Z}/p^n \to \mathbb{Z}/p$ induces an iso on K_0, since the kernel is nilpotent. Since S is Artinian, every finitely generated S-module has finite length and has a composition series with subquotients all isomorphic to the unique simple S-module $\mathbb{Z}/p$. $G_0(\mathbb{Z}/p) \cong \mathbb{Z}$ with generator $[\mathbb{Z}/p]$. The Cartan map $K_0(S) \to G_0(S)$ sends $[\mathbb{Z}/p^n]$ to $[\mathbb{Z}/p^n]$, which has length n and is thus equal to n times the generator $[\mathbb{Z}/p]$. So if $n > 1$, the Cartan map for S is not an isomorphism.

14.6 Dévissage

(French word, from déviser, to divide) Motto: Divide and conquer

Theorem 14.6 (Theorem II6.3 in Weibel, generalize the previous example) Let $\mathcal{B} \subseteq \mathcal{A}$ be small abelian categories. Assume

1. $\mathcal{B}$ is a full subcategory, and all quotients and subobjects (in $\mathcal{A}$) in $\mathcal{B}$ stay in $\mathcal{B}$
2. Every object A in $\mathcal{A}$ has a finite filtration $0 \subseteq A_1 \subseteq A_2 \subseteq \cdots \subseteq A_n = A$ with all subquotients A_{j+1}/A_j lying in $\mathcal{B}$

Then the inclusion $\mathcal{B} \hookrightarrow \mathcal{A}$ is an exact functor and induces an isomorphism $K_0(\mathcal{B}) \to K_0(\mathcal{A})$

Proof. The first statement follows immediately from the assumptions. Surjectivity follows from b). since $[A] = [A_1] + [A_2/A_1] + \cdots + [A_n/A_{n-1}]$ in $K_0(\mathcal{A})$. For injectivity, construct $K_0(\mathcal{A}) \to K_0(\mathcal{B})$ similarly

□

Proposition 14.7 X is a Noetherian scheme, X_{red} is the associated reduced scheme, then $X_{\text{red}} \to X$ induces an isomorphism on G_0. If $X = \text{Spec } R$, $X_{\text{red}} = \text{Spec } (R/\text{ nil } R)$, the transfer map $G_0(R/\text{ nil } R) \to G_0(R)$ is an isomorphism

Proof. Let $\mathcal{A}$ be the category of coherent $\mathcal{O}_X$-modules, $\mathcal{B}$ the subcategory of those that factor through an $\mathcal{O}_{X_{\text{red}}}$-module. Then $G_0(X_{\text{red}}) = K_0(\mathcal{B})$ and $G_0(X) = K_0(\mathcal{A})$. Every object in $\mathcal{A}$ has as finite filtration with subquotients in $\mathcal{B}$, so we can apply Devissage to conclude that $K_0(\mathcal{B}) \to K_0(\mathcal{A})$ is an isomorphism

□

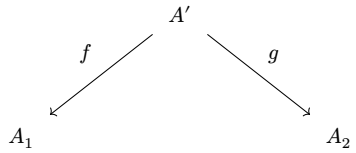
Proposition 14.8 (analogue of excision for G_0 of schemes) X Noetherian scheme, Z closed subscheme, i inclusion, $\mathcal{A}$ is the category of coherent $\mathcal{O}_X$-modules supported on Z, $\mathcal{B} =$ Coh (Z), $\mathcal{I}$ is the ideal sheaf such that $\mathcal{O}_X/\mathcal{I} = \mathcal{O}_Z$, identify $\mathcal{B}$ with subcategory of $\mathcal{A}$ annihilated by $\mathcal{I}$, then $\mathcal{B} \to \mathcal{A}$ induces an isomorphism on K_0

Proof. An object $\mathcal{F}$ in $\mathcal{A}$ has a finite filtration $\mathcal{F} \supset I\mathcal{F} \supset I^2F \supset \cdots$ with subquotients in $\mathcal{B}$

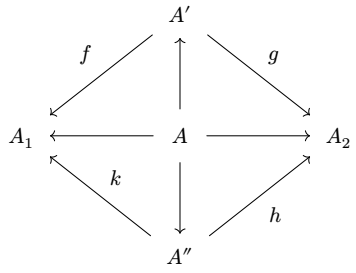
□

14.7 Localization

Let $\mathcal{A}$ be a small ableian category, $\mathcal{B}$ a **Serre subcategory**, i.e. a subcategory closed under subobjects, quotients and extensions. Define the **quotient category** $\mathcal{A}/\mathcal{B}$ as follows. The objects are the same as objects of $\mathcal{A}$, A morphism $A_1 \to A_2$ is an equivalence of classes of diagrams



with f an isomorphism mod $\mathcal{B}$ (i.e. $\ker f, \operatorname{coker} f$ lie in $\mathcal{B}$). Two such diagrams are equivalent if one has a commutative diagram



With $A' \leftarrow A \to A''$ isomorphisms mod $\mathcal{B}$

This construction is universal so that there is an exact **quotient functor** $q : \mathcal{A} \to \mathcal{A}/\mathcal{B}$, and if $T : \mathcal{A} \to \mathcal{C}$ is an exact functor, $T(\mathcal{B}) = 0$, then T factors uniquely through $q : \mathcal{A} \to \mathcal{A}/\mathcal{B}$

Example 14.9 $\mathcal{A}$ is the category of finitely generated abelian groups chosen in small skeleton, $\mathcal{B}$ is the category of p-primary torsion groups

Theorem 14.10 (Localization theorem) $\mathcal{A}$ small abelian category, $\mathcal{B}$ Serre subcategory, i.e. a subcategory closed under subobjects, quotients and extensions. Then the sequence $K_0(\mathcal{B}) \xrightarrow{i_*} K_0(\mathcal{A}) \xrightarrow{q_*} K_0(\mathcal{A}/\mathcal{B}) \to 0$ is exact

Proof. $q_* : K_0(\mathcal{A}) \to K_0(\mathcal{A}/\mathcal{B})$ is surjective, and the composite $K_0(\mathcal{B}) \to K_0(\mathcal{A}) \to K_0(\mathcal{A}/\mathcal{B})$ is 0 in $K_0(\mathcal{A}/\mathcal{B})$, since we have an isomorphism between $\mathcal{B}$ and 0 in $\mathcal{A}/\mathcal{B}$ given by showing that $B \xrightarrow{0} B$ and $B \xrightarrow{\mathrm{id}} B$ are equivalent

Suppose $[A_1] = [A_2] \in K_0(\mathcal{A})$ goes to 0 in $K_0(\mathcal{A}/\mathcal{B})$, thus $[A_1] \cong [A_2]$ in $K_0(\mathcal{A}/\mathcal{B})$, we want to show $[A_1] - [A_2] = [B_1] - [B_2]$ with B_j in $\mathcal{B}$. Suppose you have $f : A_1 \to A_2$ with kernel, cokernel in $\mathcal{B}$, so we have an exact sequence $0 \to B_1 \to A_1 \xrightarrow{f} A_2 \to B_2 \to 0$

□

Corollary 14.11 R is a Dedekind domain with field of fractions F. The the localization sequence $\bigoplus_{0 \neq \mathfrak{p} \text{ prime}} K_0(R/\mathfrak{p}) \to K_0(R) \to K_0(F) \to 0$ is exact

Proof. $\mathcal{A}$ is the category of finitely generated R-modules. By resolution theorem which we will prove later, the Cartan map $K_0(R) \to G_0(R) = K_0(\mathcal{A})$ is an isomorphism. Let $\mathcal{B}$ the subcategory of finitely generated torsion modules, and M in $\mathcal{B}$ has a primary decomposition $M = \bigoplus_{\mathfrak{p}} M[\mathfrak{p}]$, $M[\mathfrak{p}]$ is the $\mathfrak{p}$-torsion part. For fixed $\mathfrak{p}$, $K_0(R/\mathfrak{p}) \to G_0(R/\mathfrak{p})$ is an isomorphism (since $R/\mathfrak{p}$ is a field), then Devissage.

□

Proposition 14.12 R commutative Noetherian ring, $\mathfrak{p}$ is a prime ideal, then we get an exact sequence $G_0(R/\mathfrak{p}) \to G_0(R) \to G_0(R_{\mathfrak{p}}) \to 0$

Proof. Let $\mathcal{A}$ be a suitable small skeleton of the category of finitely generated R-modules. Let $\mathcal{B}$ be the Serre subcategory of p-primary torsion R-modules. Then $K_0(\mathcal{A}) = G_0(R)$ by definition, $K_0(\mathcal{B}) = G_0(R/\mathfrak{p})$ by Devissage. We claim $G_0(R_\mathfrak{p}) = K_0(\mathcal{A}/\mathcal{B})$. This is because elements of $\mathcal{A}/\mathcal{B}$ are really R-modules in which maps of modules with cokernel and kernel in $\mathcal{B}$ are declared to be isomorphisms. This effectively means that we are inverting everything not in $\mathfrak{p}$

□

14.8 $\mathbb{A}^1$-homotopy invariance

If R is a Noetherian ring, $G_0(R) \cong G_0(R[t])$. Similarly, if X is a Noetherian scheme, $G_0(X) \cong G_0(\mathbb{A}^1_X)$(or if X is a scheme over Spec k, $G_0(X) = G_0(X \times A^1)$), also $G_0(R[t]) \cong G_0(R[t, t^{-1}])$, etc.

Proof. We only do the case of rings, the case of schemes is analogous. Let $\mathcal{A}$ be the category of finitely generated $R[t]$-modules, $\mathcal{B}$ the subcategory where t acts nilpotently, by Devissage, $K_0(\mathcal{B}) = G_0(R[t]/(t)) = G_0(R)$. The quotient cat $\mathcal{A}/\mathcal{B}$ is equivalent to the category where t acts by an isomorphism. So we get the exact sequence

$$G_0(R) \xrightarrow{\pi^*} G_0(R[t]) \xrightarrow{j_*} G_0(R[t, t^{-1}]) \to 0$$

the first map π^* is the dual to the projection $\pi : R[t] \to R$, sending t to 0 (it's the transfer map viewing an R-module as a n $R[t]$-module with t acting by 0), and the second map j_* is induced by the flat inclusion $R[t] \hookrightarrow R[t, t^{-1}]$, π^* is 0 since if M is finitely generated R-module, we have the exact sequence of $R[t]$-modules

$$0 \to M[t] \xrightarrow{t} M[t] \to M \to 0$$

giving(in $G_0(R[t])$), $[M] + [M[t]] = [M[t]]$, i.e. $[M] = 0$, thus $G_0(R[t]) \cong G_0(R[t, t^{-1}])$

□

To sum up, we've proved the

Theorem 14.13 (Fundamental theorem for G_0) R Noetherian ring, Then $G_0(R) \underset{i_*}{\overset{\cong}{\to}} G_0(R[t]) \underset{j_*}{\overset{\cong}{\to}} G_0(R[t, t^{-1}])$. The corresponding statement for K_0 is **false**, for two reasons

1. $K_0(R[t])$ can be bigger than $K_0(R)$ if R is not regular. One has $K_0(R[t]) \cong K_0(R) \oplus NK_0(R)$
2. When R isn't regular, there is another contribution to $K_0(R[t, t^{-1}])$ called $K_{-1})(R)$ (This is in addition to the cokernel $NK_0(R)$ of $K_0(R) \to K_0(R[t])$), which occurs twice, once because of $K_0(R[t^{-1}])$.

14.9 Euler characteristics and the resolution theorem

Now we want to come back to the Cartan map $c : K_0(R) \to G_0(R)$ for R a Noetherian ring.

Theorem 14.14 (Resolution theorem) R Noetherian ring of **finite projective dimension**, i.e., such that any finitely generated R-module M has a finite resolution

$$0 \to P_n \to P_{n-1} \to \cdots \to P_0 \to M \to 0$$

by finitely generated projective R-modules (the scheme is smooth and reduced), then the Cartan map $c : K_0(R) \to G_0(R)$ is an isomorphism, and $[M] = c([P_0] - [P_1] + \cdots \pm [P_n])$. This is related to the **Euler characteristic** of a resolution, which is $[P_0] - [P_1] + \cdots \pm [P_n]$. These Euler characteristics also show up in the **Wall finiteness obstruction**, which is a necessary and sufficient for a connected **finitely dominated** CW-complex to be homotopy equivalent to a finite CW-complex

Proof.

□

Definition 14.15 Let X be a CW-complex. We say X is **finitely dominated** if there is a finite CW-complex Y and there are maps



such that $r \circ i \simeq \mathrm{id}_X$.

Obviously if X is a retract Y, i.e., we can choose i injective and r so that $r \circ i = \mathrm{id}_X$, then X is finitey dominated.

The interest in this condition is that certain general topology conditions give this for free (see theory of ANR's, absolute neighborhood retracts.)

Wall's idea: Suppose X is dominated as above, then you can reduce to the case where X is connected, and $C_*(\tilde{X})$, the cellular chain complex of $\tilde{X}$, sits as a direct summand in $C_*(\tilde{Y})$, as complexes of $\mathbb{Z}\pi_1(X)$-modules. Since Y is finite, $C_*(\tilde{Y})$ is finite generated free over $\mathbb{Z}\pi$. So $C_*(\tilde{X})$ are a chain complex of finitely generated projective $\mathbb{Z}\pi$-modules.

If X is finite up to homotopy, we can choose a model for X (up to homotopy) in which $C_*(\tilde{X})$ is finitely generated and free.

The **Wall finiteness obstruction** is the alternating sum $\sum_{i=0}^{\dim X}(-1)^i[C_i(\tilde{X})]$ in $\tilde{K}_0(\mathbb{Z}\pi) = K_0(\mathbb{Z}\pi)/$ copy of $\mathbb{Z}$ coming from free modules

Key idea: Suppose R is a ring, M is an R-module with a finite projective resolution, or more generally, any chain complex of f.g. projective R-modules. Then the Euler characteristic $\sum(-1)^i[P_i]$ in $K_0(R)$ is in the first case, an invariant of M in the second case, an invariant chain homotopy equivalent to M

Proof. For surjectivity, split up the resolutions to exact sequences, and add up to get the Euler characteristic = [M]

$$0 \to P_n \overset{\alpha_n}{\to\to} \cdots \to P_0 \overset{\alpha_0}{\to\to} M \to 0$$

For injectivity, want to show χ(M) is independent of the choice of the resolution. Since any two resolutions are chain homotopic, construct the cone of two resolutions $P_\bullet = P'_\bullet \oplus P''_\bullet$, have short exact sequence $0 \to P' \to P \to P''_{\bullet+1} \to 0$, Euler characteristic is additive, $\chi(P) = \chi(P') - \chi(P'')$, but since P is acyclic, so $\chi(P) = 0$

□

Exact categories An exact cat is a full subcat containint 0 and closed under extensions

Example 14.16 A cat of f.g. R module C cat of f.g. proj R module, C is a exact cat in abelian cat A

Definition 14.17 $\mathcal{C} \subseteq \mathcal{A}$ is a an exact category of a small abelian category $\mathcal{A}$, then we define $K_0(\mathcal{C})$ to be the free abelian group on objects of $\mathcal{C}$ module relations $[P] = [P'] + [P'']$. And the Cartan map is $K_0(\mathcal{C}) \to K_0(\mathcal{A})$

Example 14.18 R is a left Noetherian ring, $\mathcal{A}$ a small model for f.g. R modules, C=f.g., the usual example

Example 14.19 X is a Noetherian scheme, $\mathcal{A}$ is the category of coherent $\mathcal{O}_X$-modules, $\mathcal{C}$ is the category of locally free coherent $\mathcal{O}_X$ modules, i.e. the category of algebraic vector bundles over X, then $K_0(\mathcal{A}) = G_0(X)$, $\mathcal{K}_0(\mathcal{C}) = K_0(X)$

Example 14.20 R is a ring(not necessarily Noetherian). Let $\mathcal{A}$ be a small model for the category pseudo-coherent R-modules, i.e. ones with resolution by f.g. proj R moduels, whrere the resoluiton dones't have to be finite. If R is Noetherian, every finite generated R-module is pseudo-coherent. Then we define $G_0(R) = K_0(\mathcal{A})$, and we get a Cartan map $K_0(R) \to G_0(R)$. And analogue for non-Noetherian schemes

Example 14.21 R is a ring, define Nil (R) to be the category of pairs (P, T) with P a finitely generated(right) R-module(in a small model), and $T : P \to P$ is a nilpotent R-module endomorphism. Morphisms are commutative diagrams

$$\begin{array}{ccc} P & \xrightarrow{T} & P \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f} \\ P' & \xrightarrow[T']{} & P' \end{array}$$

This is an exact subcategory of the larger category of pairs (P, α), with α not required to be nilpotent If R is right Noetherian, there is a still larger abelian category of f.g. right R-modules

What is this good for? Recall that $G_0(R[t]) \cong G_0(R)$ for R Noetherian, and the analogue fails for K_0, K_0(Nil R) is the missing piece in $K_0(R[t])$, i.e. $K_0(R[t]) \cong K_0$(Nil (R)). There is an obvius summand $K_0(R)$ inside K_0(Nil (R)) given by $P \mapsto (P, 0)$ and $(P, T) \mapsto P$. Sometimes K_0(Nil (R)) $\cong K_0(R)$, sometimes no

Recall there is an exact sequence

$$\mathrm{GL}_(R) \to \mathrm{GL}\ (R/I) \to K_0(I) \to K_0(R) \to K_0(R/I)$$

This suggests that $K_1(R)$ should be related to GL(R), but homology groups shooulde be commutative

Guess K_1(R)≅GL(R)_ab

Theorem 14.22 R is a Euclidean domain, then $K_1(R) \cong R^\times$

Proof.

□

Remark 14.23 The determinant $\det : \mathrm{GL}_n(R) \to R^\times$ givs split surjection $K_1(R) \to R^\times$, the kernel is denoted $SK_1(R)$, $SK_1(R) = 0$ for euclidean domain, but not generally for PID

Theorem 14.24 (Dieudonné) R is a division ring, there is a unique determinant map giving an isomorphism $K_1(R) \to R^\times_{ab}$

Example 14.25 $\mathbb{H}$ is the quaternions, $\mathbb{H}^\times \cong \mathbb{R}^\times_+ \times SU(2)$, since $SU(2)$ is simple, $K_1(\mathbb{H}) \cong \mathbb{R}^\times_+$

Proof. There is a unique map such tat $det(EA = det(A)$ detinduces an iso $K_1(R) \to R^\times_{ab}$, we have split surjection $\mathrm{GL}_n(R)_{ab} \to R^\times_{ab}$ with the splitting map induced by inclusion into the first component

□

D division ring center F, dim_FD=n^2, n is the dim over F of a maximal commutative subfield. say E

Example 14.26 D=H^2, F=R,E=C, then n=2,dim_FD=2^2=4,dim_ED=2

Theorem 14.27 N_{red} is an iso

D↪M_n(E) by tensoring with E, $D^\times \hookrightarrow \mathrm{GL}_n(E)$, so the det map $GL_n(E) \to E^\times$ induces $D^\times \to E^\times$. If E/F is Galois, then Gal(E/F), get N_red:K_1(D)=D^×→F^×

Example 14.28 D=H^2, F=R,E=C, the reduced norm is a multiplicative map $D^\times \to F^\times$. On $F^\times$, for a∈F^×, a↦det(a,⋯,a)=a^n, sending i,j,k to 1,so N_red(z=a+bi+cj+dk)=z|z

Proof. Want to show N_{red} is an iso mod n. In general, it's neither injective nor surjective, but the kernel and cokernel are n-torsion. This map almost split by the inclusion $F^\times \hookrightarrow D^\times$. So coker N_red is a quotient of $F^\times/(F^\times)^n$, can similarly check that $\ker N_{red}$ is n-torsion

□

The Whitehead Group $Wh(G)$

GL_1(ZG)=(ZG)^×

Definition 14.29 Wh(G)=K_1(ZG)/(Z^××G)

$K_0(R \text{ rel } S)$ to be K_0 of the Waldhausen category of bounded complexes of finitely generated projective R-modules P_•such that S^-1P is exact

Theorem 14.30 R=k[t], K_1(R)≅k[t]^×≅k^×, S=t,t^2,⋯, S^-1R=k[t,t^-1] is a PID, K_0(R rel S) is generated by [k[t]→tk[t]]. The exact sequence becomes

$$\begin{array}{ccccccccc} K_1(R) & \longrightarrow & K_1(S^{-1}R) & \longrightarrow & K_0(RrelS) & \longrightarrow & K_0(R) & \longrightarrow & K_0(S^{-1}R) \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ k^\times & \longrightarrow & k^\times \times \mathbb{Z} & \longrightarrow & \mathbb{Z} & \xrightarrow{0} & \mathbb{Z} & \xrightarrow{\cong} & \mathbb{Z} \end{array}$$

Fundamental theorem R is a ring, consider $R[t]$, $S = \{t, t^2, \cdots\}$, $S^{-1}R[t] = R[t, t^{-1}]$. We want to relate K-theory of R, $R[t]$ and $R[t, t^{-1}]$. Let Nil (R) be the category of pairs (P, T) where P is a finitely generated projective R-module, $T \in \text{End } (P)$ nilpotent. This is an exact category with maps commutating diagrams

The category of finitely generated projective R-modules embeds in Nil (R) as the pairs $(P, 0)$. So we have a split injection $K_0(R) \to K_0(\text{Nil } (R))$. Note that for "nice" rings, e.g. regular Noetherian rings, this map is an isomorphism and $K_0(\text{Nil } (R)) \cong K_0(R)$

Recall the localization theorem gives an exact sequence $K_1(R[t]) \to K_1(R[t, t^{-1}]) \xrightarrow{\partial} K_0(R[t] rel\{t,^2, \cdots\}) \xrightarrow{0} K_0(R[t]) \hookrightarrow K_0(R[t, t^{-1}])$

It will turn out that $K_1(R[t]) \cong K_1(R) \oplus \text{coker } (K_0(R) \to K_0(\text{Nil } (R))) =: K_1(R) \oplus NK_1(R)$. Similarly, we have $K_0(R[t]) \cong K_0(R) \oplus NK_0(R)$. If we had the algebraic analogue of homotopy invariance, NK_j would be 0

Lemma 14.31 g∈GL(R[t]) is conjruent ot the identity mod t, then [g]=[1-νt] in K_1(R[t]) for some nilpotent $\nu \in M_n(R)$(n maybe large compare to the size of g)

Proof. Think of g as a poly in t with coef that are matices over R, the constant term being the identity matrix, if the polynomial is of deg 1, then g=1-νt, but g being invertible, $g^{-1} = 1 + \nu t + \cdots$ has to terminate, so $\nu$is nilpotent. Induction for degg, so assume g=1-νt+γt^n for some nilpotent ν,

$$\begin{array}{c} g \\ \&1 \sim g\&\gamma t\text{^}n\text{-}1 \\ \&1 \sim 1\text{-}\nu t\&\gamma t\text{^}n\text{-}1 \\ \text{-}t1\&1 = 1 \\ \&1\text{-}t\ \nu\&\gamma t\text{^}n\text{-}2 \\ 1\&0 \end{array}$$

□

Theorem 14.32 K_1(R[t])≅K_1(R)⊕NK_1(R), where $NK_1R \oplus K_0(R) \cong K_0(NilR)$

Proof. Map ker(K_1(R[t]→K_1(R))) NK_1(R) to use lemma. For νnilpotent, send to 1-νt∈ker (K_1(R[t]→K_1(R)))

□

Theorem 14.33 (Fundamental theorem of K-theory) $K_1(R[t,t^{-1}]) \cong K_1(R) \oplus NK_1(R) \oplus NK_1(R) \oplus K_0(R)$, ($NK_1(R)$ from $R[t]$ and $R[t^{-1}]$)

Proof. Construct exact seq 0→K_1(R)→ΔK_1(R[t])+K_1(R[t^-1])→(a,b)↦a-b K_1(R[t,t^-1])→∂K_0(R)→ 0 ∂is split by the map [P]→t
&1 Localization theorem gives K_1(R[t])→K_1(R[t,t^-1])→∂K_0(R[t]rel S)→K_0(R[t])→ K_0(R[t,t^-1]) the last map is injective, so we have exact seq K_1(R[t])→K_1(R[t,t^-1])→∂ K_0(R[t]rel S)→ 0

□

Lemma 14.34 K_0(R[t]rel S)≅K_0(Nil(R)), which by the last theorem is K_0(R)⊕NK_1(R)

Proof. K_0(R[t]rel S) is generated by length one complexes R[t]^n→αR[t]^n, wehre αis iso after inverting t. If P is a f.g. proj R module, Po+Q≅R^n, then P∈K_0(R) can be identityfed with R[t]^n→R[t]^n which is the identity on Q[t] and mult by t on P[t] Similarly, given nu nil, t1-nu is an iso after inverting t, sicne (t1-nu)^-1t^-1(1-nu t)^-1∈GL_n(R[t,t^-1]), so repre a class in K_0(R[t]rel S) A gneral S iso R[t]^n→αR[t]^n is given (since t is not a zero divisor) by an exact seq 0→R[t]^n→αR[t]^n→M→ 0 M is annilated by some power t^m of t, tensor the seq with R[t]/(t^m), we obtain 0→Tor_1^R[t](M,R[t]/(t^m))→(R[t]/(t^m))^n→(R[t]/(t^m))^n→ M→ 0 But from the free resolution 0→R[t]→t^m R[t]→R[t]/(t^m)→ 0

□

$$[P,T] \mapsto [(P \oplus Q), 1 - Tt^{-1}$$

Construct splitting $K_0(\text{Nil }(R)) \to K_1(R[t,t^{-1}])$, $1]$, $P \oplus Q \cong R^n$

Proposition 14.35 R is an algebra over F_p, then NK_1(R) is a p group

Proof. Consider $R = F_p[x]/(x^n)$, (1+tf(x,t))^p^m=1+t^p^mf(x^p^m,t^p^m)

□

Proposition 14.36 R algebra over Q, then NK_1(R) is a Q vector space

Proof. Similar argument to reduce to $R = Q[x]/(x^n)$. Use Witt vectors

□

Definition 14.37 F : rings → Groups is a functor. Let $LF(R) = \text{coker } (F(R[t] \oplus F(R[t^{-1}])) \to F(R[t,t^{-1}]))$ Then we get a sequence Seq(F,R)

$$0 \to F(R) \stackrel{\Delta}{\to} F(R[t]) \oplus F(R[t^{-1}]) \to F(R[t,t^{-1}])) \to LF(R) \to 0$$

F is acyclic if

The fundamental theorem syas K_1 is a contracted funcor with $LK_1 = K_0$ Defninte $K_n = LK_{1-n}$, $NF(R) = \text{coker } (F(R) \to F(R[t]))$ If is regular and Noe, then the resolution theorme showed $K_0(R) = G_0(R)$ and $NK_0(R) = 0, K_{-1}(R) = 0$. In fact $K_{-n}(R) = 0 \forall n \geq 1$. So negative K theory is mostly interesting for non-regular rings

Example 14.38 $R = \mathbb{Z}$, $I = (n)$, $R/I = \mathbb{Z}/n$. If n has more than one prime factor, $Z/n \cong Z/p_1^{r_1} \oplus \cdots \oplus Z/p_k^{r_k}$ a product of k local rings, $K_0(R/I) = Z^k$, $K_0(R) = Z$. We will obtain a long exact sequence

Proposition 14.39 If F is contracted functor, then so are LF,NF and $NLF \cong LNF$

Proof.

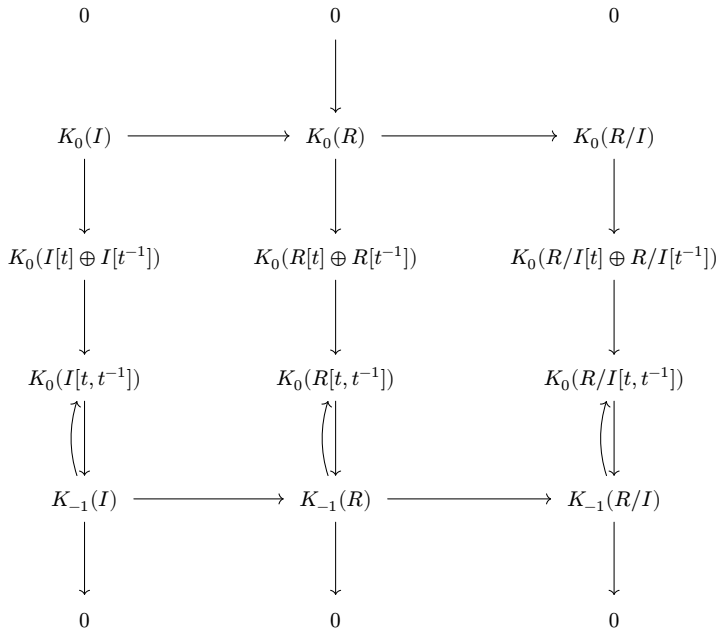
□

Theorem 14.40 (Fundamental theorem) $K_n(R[t,t^{-1}]) \cong K_n(R) \oplus K_{n-1}(R) \oplus NK_n(R) \oplus NK_n(R)$, for $n \leq 1$

Corollary 14.41 If F is a contracted functor, then $F(R[t_1,\cdots,t_n]) \cong (1+N)^n F(R)$, $F(R[t_1^{\pm 1},\cdots t_n^{\pm 1}]) \cong (1+2N+L)^n F(R)$

Theorem 14.42

Proof. Use the definition of LK_0 to get



We get a boundary map $\partial : K_0(R/I) \to K_{-1}(I)$, but this comes from the boundary map $K_1(R/I[t,t^{-1}]) \xrightarrow{\partial} K_0(I[t,t^{-1}])$

□

Definition 14.43 A ring R is called K_n regular for some n if LK_n, NK_n both vanish. A regular Noetherian ring is K_n regular for all n

Fact 14.44 K_1 regular $\Rightarrow$ K_0 regular $\Rightarrow$ K_{-1} regular $\Rightarrow$, None of these arrows are reversible. In general, NK groups and lower K groups can often be computed using the definition and polynomial, Laurent polynomial rings. NK groups can also be computed using the Nil category. Recall $NK_1(R) = \ker(K_0(\text{Nil }(R)) \to K_0(R))$ Applying the machinery of contracted functors, you can also get a description of NK_0. There are lots o interesting contracted functors, especially on commutative rings, other than K-theory functors

Example 14.45 The group of units $U(R)$ is a contracted functor, Pic is a contracted functor

Still another approach to negative K-theory(due to Karoubi). Show that $K_n(R) = K_n(M(R))$, where $M(R) = \lim M_k(R)$(Morita invariance) clearly true for K_0, K_1. This inherits to LK_n, NK_n. Embed $M(R)$ as an ideal in a larger ring which has identically 0 K-groups. From the exact sequence $0 \to M(R) \to B(R) \to Q(R) = B(R)/M(R) \to 0$

There is an obvious map $\text{St}_n(R) \to E_n(R)$

Lemma 14.46 The subgroup of $\text{St}_n(R)$ generated by $x_{ij}(r), i<j$, maps iso to the group of strictly upper triangular matrices in $\text{GL}_n(R)$, $n \geq 3$

Definition 14.47 1→A→E→G→ 1 is a central extension if A is abelian and central in E

Theorem 14.48 A group G has a universal central extension E iff G is perfect iff E is perfect and all central extensions of E are trivial

Proof. Consider presentation $1 \to \mathrm{R} \to \mathrm{F} \to \mathrm{G} \to 1$ F is free, then $G = [F, F]/R$, let $E = [F, F]/[F, R]$, we have $E \to G$, the kernel is central, E is perfect

□

Theorem 14.49 (Milnor) $\mathrm{St}_n(R)$ is the universal central extension of $E(R)$. Define $K_2(R)$ to be the kernel of $\mathrm{St}\ (R) \to E(R)$

Proof. St (R) is perfect

□

Corollary 14.50 $K_2(R)$ is naturally isomorphic to $H_2(E(R), \mathbb{Z})$

Theorem 14.51 G is a group, G has a universal central extension if $H_1(G, \mathbb{Z}) = 0$

Definition 14.52 A *Waldhausen category* is a small(skeleton small) category $\mathcal{C}$ with two special classes of morphisms: cofibrations $A \rightarrowtail B$, weak equivalences $A \to B$, satisfying:

1. Every iso is both a cofibration and a weak equivalence
2. There is a zero object, and for every $A \in \mathcal{C}$, $0 \rightarrowtail A$ is a cofibration
3. the pushout of a cofibration along any morphism exists and is a cofibration, i.e.

$$\begin{array}{ccc} A & \rightarrowtail & B \\ \downarrow & & \downarrow \\ C & \rightarrowtail & B \cup_A C \end{array}$$

   Note that if $A = 0$, $B \cup_0 C = B \coprod C$. Also the pushout square

$$\begin{array}{ccc} A & \rightarrowtail & B \\ \downarrow & & \downarrow \\ 0 & \rightarrowtail & B \cup_A 0 \end{array}$$

   gives us the cokernel B/A
4. Weak equivalences are closed under compositions, and we can glue weak equivalences

$$\begin{array}{ccccc} C & \longleftarrow & A & \longrightarrow & B \\ \simeq\downarrow & & \simeq\downarrow & & \simeq\downarrow \\ C' & \longleftarrow & A' & \longrightarrow & B' \end{array}$$

Definition 14.53 Let $\mathcal{C}$ is a Waldhausen category, then $K_0(\mathcal{C})$ is the free abelian group on the objects modulo relations

1. $[A] = [A']$ if there is a weak equivalence $A \xrightarrow{\simeq} A'$
2. $[C] = [B] + [C/B]$ if there is a cofibration sequence $B \rightarrowtail C \to C/B$. One gets the relation $[0] = 0$, $[B \cup_A C] = [B] + [C] - [A]$ as a consequence

Example 14.54 $\mathcal{C} \subseteq \mathcal{A}$ is an exact category, then $\mathcal{C}$ is a a Waldhausen category, with cofibration sequences the short exact sequence

Example 14.55 $\mathcal{C}$ is the category of based finite CW complexes, $0 = *$ is the zero object, $0 \to A$ is the inclusion, cofibrations are based inclusions of subcomplexes, weak equivalences are based homotopy equivalences

Proposition 14.56 (Waldhausen) $K_0(\mathcal{C}) = \mathbb{Z}$, given by the Euler characteristic $A \to \chi(A)$

Proof. $S^{n-1} \rightarrowtail D^n \to S^n$, thus $[S^{n-1}] + [S^n] = 0$, so $[S^0]$ generates $K_0(C)$, identify this with $1 \in \mathbb{Z}$, then $[S^n] = (-1)^n$

□

Example 14.57 A small abelian category, $\mathcal{C}$ the category of bounded chain complexes, $\mathcal{C}$ is Waldhausen, cofibrations are monic chain maps and weak equivalences are quasi isomorphisms

Theorem 14.58 (SGA) In the situation of Example 14.55, the map $A_* \to \chi(A_*)$ is an iso $K_0(\mathcal{C}) \to K_0(\mathcal{A})$

Proof. By "Euler-Poincare principle", $\chi(A_*) = \chi(H_*(A_*))$, $0 \to Z_n \to A_n \to A_{n-1} \to A_{n-1}/B_{n-1} \to 0$, $0 \to B_n \to Z_n \to H_n \to 0$, and add these. The Euler characteristic is also additive on cofibration sequences $B_* \rightarrowtail A_* \to A_*/B_*$

□

Theorem 14.59 (Approximation theorem) $F : \mathcal{A} \to \mathcal{B}$ is exact(take cofibrations to cofibrations) between Waldhausen categories. Assume

1. f is a weak equivalence in $\mathcal{A}$ iff $F(f)$ is a weak equivalence in $\mathcal{B}$
2. Any $b : F(\mathcal{A}) \to \mathcal{B}$ factors as $b'F(a')$, $b' : B' \to B$ is a weak equivalence, $a : A \to A'$ is a cofibration with $F(A') = B'$
3. can choose skeleton small?

Then F induces iso $K_0(\mathcal{A}) \to K_0(\mathcal{B})$

Proof. Consider $A = 0$, we see that any $B \in \mathcal{B}$ is weakly equivalent to $F(A')$, thus $K_0(\mathcal{A}) \to K_0(\mathcal{B})$ is surjective. The sets of weak equivalence classes in $\mathcal{A}$ and in $\mathcal{B}$ coincide, or in bijection under F_*

□

Application: Let R be a ring, we have the following Waldhausen categories:

- Proj (R), finite generated projective R moudles
- $\mathrm{Ch}^b(\text{Proj } (R))$, bounded complexes of finitely generated R modules
- Perfect chain complexes, i.e. chain complexes quasi isomorphic to complexes in $\mathrm{Ch}^b(\text{Proj } (R))$

The Approximation theorem says they have the same K_0, namely $K_0(R)$

The real use of Waldhausen categories will be in the construction of higher K-theory, where Waldhausen "S-construction" gives a simple elegant approach covering all of these examples

Example 14.60 Fix a field k, say $\mathbb{C}$, $\mathcal{A}$ be the category of smooth quasi-projective varieties over k. This is a Waldhausen category with cofibrations being closed inclusions, weak equivalences being isomorphisms, $[\mathbb{P}^1] = [*] + [\mathbb{A}^1]$

Example 14.61 R is a ring, let End (R) be the category of pairs (P, T), P is a finitely generated projective R-module, $T : P \to P$, morphisms are commutative diagrams

$$\begin{array}{ccc} P & \longrightarrow & P' \\ {\scriptstyle T}\downarrow & & \downarrow{\scriptstyle T'} \\ P & \longrightarrow & P' \end{array}$$

You get $K_0(\text{End } (R))$ containing $K_0(R)$ as a direct summand via the functor $P \mapsto (P, 0)$. Compute the rest of $K_0(\text{End } (R))$. This was studied by Grayson

15 Introduction to higher K theory

Now that we've seen an **ad hoc** construction of K_0, K_1, K_2, and negative K-groups, it is time to study how one could get all the K-groups at once, so that they fit together as that they fit together as a unit. Quillen's idea, which won him the Fields medal in 1972, was to construct not a sequence of groups, but a **topological space**, or actually a **spectrum**, so that K_i is just the i-homotopy group of this space. In other words, out of a ring R (or later, an abelian category or exact category or Waldhausen category) we want a space $K(R)$ so that $K_i(R) = \pi_i(K(R))$. Of course, certain key conditinos have to be satisfied:

1. The association $R \mapsto K(R)$ should be functorial.
2. Given an ideal $I \subseteq R$, we should get a fibration $K(R, I) \to K(R) \to K(R/I)$, so that the induced long exact sequence of homotopy groups has the form
$$\cdots \to K_i(R, I) \to K_i(R) \to K_i(R/I) \xrightarrow{\partial} K_{i-1}(R, I) \to \cdots$$
3. The groups K_0, K_1, K_2 should agree with the ones we've already defined. To motivate the simplest construction, the Quillen +-construction, recall that topological K-theory K^{top} is given by homotopy classes of maps $X \to K_0(\mathbb{C}) \times B \text{ GL } (\mathbb{C})^{\text{top}}$

Theorem 15.1

$$K_i(F_q) = \begin{cases} Z, i = 0 \\ \text{cyclic of order } \ q^k - 1, i = 2k - 1 \\ 0, i \ \text{ even} \geq 2 \end{cases}$$

More precisely, $K(F_q)$ is homotopy equivalent to the homotopy fiber of the map $BU \xrightarrow{\psi^q - 1} BU$, where ψ^q is the Adams operation

Idea: Given a matrix with entries in $\bar{F}_q$, which is roughly analogues to $\mathbb{C}$, it has entries in F_q iff it is fixed under Frobenius map. Once you make this precise, $K_i(F_q)$ falls out from the homotopy sequence since $\pi_i(BU) = \begin{cases} Z, i \text{ even} \\ 0, i \text{ odd} \end{cases}$, and $\psi^q - 1$ on π_{2k} acts by q^k, and then consider the long exact sequence of this fibration

15.1 Motivation

We've talked a lot about defining K-theory. But how do you actually compute it in practice? For motivation, it helps to look at the easier case of topological K-theory. There we had characteristic classes which provide a lot of information. We also had the splitting principle, which, you may recall, says that "one can pretend every vector bundle is a direct sum of line bundles." Using these, one can define the Chern character $\text{Ch} : K^0(X) \to \oplus_i H^{2i}(X, \mathbb{Q})$ for X a finite CW-complex by $\text{Ch } (L_1 \oplus \cdots \oplus L_n) = \sum_j e^{c_1(L_j)}$, for $L_1, \cdots, L_n$ line bundles with characteristic classes $c_1(L_j) \in H^2(X, \mathbb{Z})$ and with

$$e^{c_1(L_j)} = 1 + c_1(L_j) + \frac{1}{2} c_1(L_j)^2 + \cdots$$

The sum is finite since $c_1(L_j)$ is nilpotent. The splitting principle and naturality imply that Ch is well-defined for all vector bundles, and in fact for all formal differences of vector bundles, since we can define $\text{Ch } (E - F) = \text{Ch } (E) - \text{Ch } (F)$

Theorem 15.2 Ch is a ring homomorphism $K^0(X) \to H^{\text{even}}(X, \mathbb{Q})$ and its kernel is a finite nilpotent ideal. Furthermore, by Bott periodicity, it extends to a homomorphism of graded rings $K^*(X) \to H^*(X, \mathbb{Q})$, with finite kernel. It's an isomorphism after tensoring $K^*(X)$ with $\mathbb{Q}$. So cohomology almost completely computes topological K-theory. And there is a spectral sequence $H^p(X, K^q(\text{pt}) \Rightarrow K^{p+q}(X)$. This suggests that having the right cohomology theories should give most of the information about algebraic K-theory as well, and indeed this is the case. There are two main theories that apply:

- Cyclic homology and its variants, especially topological cyclic homology TC
- Motivic (co)homology for schemes

15.2 Motivic cohomology

For X a smooth quasi-projective scheme over a field k, there are complexes of sheaves $\mathbb{Z}(j), j \geq 0$ over X and the motivic cohomology of X is $H^{n,j}(X, \mathbb{Z})$ is defined to be $H^n(X, \mathbb{Z}(j))$. Note the bigrading. These groups are only nonzero for $j \geq 0$, and $\mathbb{Z}(0) = \mathbb{Z}$, $\mathbb{Z}(1) = \mathbb{G}_m[-1]$. They are related to the usual Chow groups by $\text{CH}^i(X) = H^{2i}(X, \mathbb{Z}(i))$. There is a spectral sequence (for q even)

$$E_2^{p,q} = H^p(X, \mathbb{Z}(-q/2)) \Rightarrow K_{-p-q}(X)$$

For $X = \text{Spec } k$, $H^{j,j}(X) = K_j^M(k)$, the Milnor K-theory ($\mathbb{Z}$ for $j = 0$ and generated by $k^\times$ with the Steinberg relations for $j \geq 1$). So all of this gives considerable information about K-theory of fields. There is also the Suslin-Nestorenko Theorem: $K_n^M(k) \to K_n(k)$ followed by the Chern class $c_{n,n} : K_n(k) \to H^{n,n}(\text{Spec } k, \mathbb{Z}) = K_n^M(k)$ is multiplication by $(-1)^{n-1}(n-1)!$, and in particular, is an injection modulo torsion of order involving only primes $\leq n-1$.

Motivic cohomology thus turns out to be a very powerful tool for computing K-theory of schemes

Some results proved with this technology: Milnor conjecture, proved by Voevodsky, Bloch-Kato conjecture, proved by Voevodsky, Rost, et al. Information on Quillen-Lichtenbaum conjecture

16 Homeworks

16.1 Homework 1

Exercise 16.1 For R a commutative ring and P an R-module, we have P finitely generated projective $\Leftrightarrow$ for every $p \in \operatorname{Spec} R$, $\exists s \notin p$ with $P[\frac{1}{s}]$ finitely generated and free. This implies that for every $p \in \operatorname{Spec} R$, P_p is finitely generated and free, and that condition is equivalent if R is Noetherian. Show that R not Noetherian, the conditions are not equivalent, i.e., it could be that P_p is finitely generated free for every prime ideal p, but P is not finitely generated projective

Solution.

□

Exercise 16.2 Let R be a commutative ring. Show that $\operatorname{Spec} R$ (with the Zariski topology) is a connected topological space (i.e., is not the disjoint union of two non-empty open sets) $\Leftrightarrow$ the only idempotents in R are 0 and 1. (We mentioned this in class but didn't prove it.) In particular, $\operatorname{Spec} R$ is connected if R is an integral domain

Solution.

□

Exercise 16.3 Work out the details of Example I.3.10.1 in Weibel: if k is a field and $R = k[t^2, t^3]$, a subring of $k[t]$, then $\operatorname{Pic}(R) \cong k$(the additive group of the field), with the isomorphism sending $\alpha \in k^\times$ to the ideal of R generated by $I_\alpha = R \cap (t - \alpha^{-1})k[t]$ and sending $0 \in k$ to R itself(So I_α is the ideal of R gotten by intersecting R with the ideal generated by $t - \alpha^{-1}$ in $k[t]$). Since I_α is obviously an ideal, it is an R-module. Show that it is locally free of rank 1, and that all non-trivial line bundles on R are isomorphic to ones arising this way

Solution.

□

References

Charles Weibel. *The K-Book*.

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